

# **Core Aptitude for Placements**

**Quantitative, Logical, Verbal & Technical Practice**

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# Table of Contents

- Quantitative Aptitude
- Logical Reasoning
- Verbal Ability
- Technical Aptitude
- Other Sections

# Quantitative Aptitude

## Source: Averages\Averages-1

Problem on Averages-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of average and some basic logic.

Problem:

Find the average of all even numbers between 104 and 148.

Solution:

Note: When the sequence forms Arithmetic Progression then

$$\text{Average} = (\text{first term} + \text{last term}) \div 2.$$

AP (Arithmetic Progression) is a sequence of numbers in which the difference between any two consecutive terms is always the same.

The even numbers between 104 and 148 are in an arithmetic sequence:

106, 108, 110, ?, 146

In an arithmetic sequence, the average = (First term + Last term)  $\div$  2

$$\text{Average} = (106 + 146) \div 2 = 252 \div 2 = 126 ?$$

Final Answer

? The average of even numbers between 104 and 148 = 126

## Source: Averages\Averages-2

Problem on Averages-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of average and some basic logic.

Problem:

A family consist of grandparents, parents and grandchildren.

The average age of 2 grandparents is 67 years, the average age of 2 parents is 35 years, and the average age of 3 grandchildren is 6 years.

Find the average age of the whole family.

Solution:

Note: Formula of Average = Sum of observations / No. of observations.

$$\text{Total age of grandparents} = 2 \times 67 = 134$$

$$\text{Total age of parents} = 2 \times 35 = 70$$

$$\text{Total age of grandchildren} = 3 \times 6 = 18$$

$$\text{Original total age of family} = 134 + 70 + 18 = 222$$

$$\text{Total number of family members} = 2 + 2 + 3 = 7$$

Average age of family =  $222 \div 7 = 31.71$  ?

Final Answer:

? Average age of the family = 31.71 years (approx.)

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### Source: Averages\Averages-3

Problem on Averages-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of average and some basic logic.

Problem:

The average of 6 numbers is 8.

What should be the 7th number so that the new average becomes 10?

Solution:

Note: Formula of Average = Sum of observations / No. of observations.

Since average = ( sum  $\div$  number of observation ),

So, sum = (number of observation  $\times$  average )

Original total sum of 6 numbers =  $6 \times 8 = 48$

New total sum needed for 7 numbers =  $7 \times 10 = 70$

7th number =  $70 - 48 = 22$  ?

Final Answer:

? The 7th number = 22

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### Source: Averages\Averages-4

Problem on Averages-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of average and some basic logic.

Problem:

The average weight of 8 persons increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. Find the weight of the new person.

Solution:

Note: Formula of Average = Sum of observations / No. of observations.

When the average increases by 2.5 kg, the total weight of all 8 persons increases by:

Increase in sum =  $8 \times 2.5 = 20$

This extra 20 kg is because the new person is heavier than the one who left.

So,

$$\begin{aligned}
 \text{Weight of new person} &= \text{Weight of old person} + \text{Increase} \\
 &= 65 + 20 \\
 &= 85 ?
 \end{aligned}$$

Final Answer

? The weight of the new person = 85 kg.

### Source: Averages\Averages-5

Problem on Averages-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of average and some basic logic.

Problem:

The average weight of 30 students is 60 kg.

If one student leaves, the new average becomes 59 kg.

Find the weight of the student who left.

Solution:

Note: Formula of Average = Sum of observations / No. of observations.

$$\text{Original total weight} = 30 * 60 = 1800$$

Now one student left so, number of students = 29

$$\text{New total weight} = 29 * 59 = 1711$$

$$\text{Weight of the student who left} = 1800 - 1711 = 89?$$

Final Answer

?Weight of the student who left = 89 kg.

### Source: Averages\Averages-6

Problem on Averages-6 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of average and some basic logic.

Problem:

The average temperature of a week was calculated as 32°C.

Later, it was realized that the temperature of Monday was taken as 30°C instead of 16°C.

Find the correct average temperature of the week.

Solution:

Note: Formula of Average = Sum of observations / No. of observations.

$$\text{Sum of observations} = \text{Average} \times \text{No. of observations}$$

Here No. of observations is No. of days in week = 7

Average = 32

Original total sum =  $7 \times 32 = 224$

Temperature was taken as  $30^{\circ}\text{C}$  instead of  $16^{\circ}\text{C}$

Error =  $30 - 16 = 14$

It means  $14^{\circ}\text{C}$  is the error that needs to be subtracted from the sum.

Correct total sum of temperature =  $224 - 14 = 210$

Correct average =  $210 \div 7 = 30$  ?

Final Answer

? The correct average temperature of the week =  $30^{\circ}\text{C}$

### Source: Averages\Averages-7

Problem on Averages-7 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of average and some basic logic.

Problem:

The average weight of 40 children in a class is 36.2 kg.

Three more children join the class with weights 42.3 kg, 39.7 kg, and 39.5 kg.

Find the new average weight of the class.

Solution:

Note: Formula of Average = Sum of observations / No. of observations.

Sum of observations = Averages X No. of observations.

Original total sum of weight =  $40 \times 36.2 = 1448$

Now 3 children join, so their total sum of weight =  $42.3 + 39.7 + 39.5 = 121.5$

New total weight = Original total sum of weight + Sum of weight of three new children

New total weight =  $1448 + 121.5 = 1569.5$

New number of children = 43 ( 40 + 3 )

New average =  $1569.5 \div 43 = 36.5$  ?

Final Answer

? New average weight of 43 children = 36.5 kg

### Source: Partnership\Partnership-1

Problem on Partnership-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

A and B start a business.

A invests ₹12,000 and B invests ₹8,000.

At the end of one year, they earn a profit of ₹20,000.

Find each person's share in the profit.

Solution:

Note: The profit is always divided in the same ratio as the investment ratio.

A invests ₹12,000 and B invests ₹8,000.

Investment ratio = 12000 : 8000 = 3 : 2.

Since both A and B invested their money for the same time period (1 year), the profit is divided in the same ratio as the investment ratio.

A's share =  $(\frac{3}{5}) \times 20,000 = ₹12,000$ .

B's share =  $(\frac{2}{5}) \times 20,000 = ₹8,000$ .

Final Answer: A gets twelve 12,000 and B gets 8,000.

### Source: Partnership\Partnership-2

Problem on Partnership-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

P and Q buy a hotel.

P invests 6 times what Q invests.

At the end of one year, the total profit is ₹84,000.

Find P's share in the profit.

Solution:

Note: Profit is divided in the same ratio as the investments.

Let Q's investment = ₹x.

Then P's investment = ₹6x.

So, the investment ratio of P : Q = 6 : 1.

Since both invested for the same time, the profit will be divided in the ratio 6 : 1.

Total profit = ₹84,000

Total parts = 6 + 1 = 7 parts

P's share =  $(\frac{6}{7}) \times 84,000 = ₹72,000$

Final Answer: P gets ₹72,000.

**Source: Partnership\Partnership-3**

Problem on Partnership-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

P, Q, and R start a hotel.

Q and R invest equally, but P invests only 1/5th of Q's investment.

At the end of one year, the profit is ₹22,000.

Find Q's share in the profit.

Solution:

Note: The profit is always divided in the same ratio as the investment ratio.

Let Q's investment = ₹x.

Then R's investment = ₹x (since Q and R invest equally).

P's investment = (1/5) of Q =  $(1/5) \times x = ₹x/5$ .

So, investment ratio of P : Q : R respectively =  $(x/5) : x : x = 1 : 5 : 5$

Because when we divide each term by x/5, it simplifies to 1 : 5 : 5.

Total profit = ₹22,000

Total parts = 1 + 5 + 5 = 11 parts

Q's share =  $(5/11) \times 22,000 = ₹10,000$  ?

Final Answer: Q gets ₹10,000.

**Source: Partnership\Partnership-4**

Problem on Partnership-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

Rohit starts an agency by investing ₹40,000.

After 4 months, Raj joins with an investment of ₹50,000.

At the end of one year, the total profit is ₹1,10,000.

Find Raj's share in the profit.

Solution:

Note: When the time of investment is different,

we multiply each partner's Investment by the time (in months) for which it was invested.

Rest the logic is the same ? the ratio of (Investment × Time) will be the ratio in which the profit is divided.

Rohit's investment = ₹40,000 for 12 months  $= 40,000 \times 12 = 4,80,000$

Raj's investment = ₹50,000 for 8 months  $= 50,000 \times 8 = 4,00,000$

Investment Ratio of Rohit : Raj = 4,80,000 : 4,00,000 = 6 : 5



So, Profit Sharing Ratio of Rohit : Raj = 6 : 5

Total profit = ₹1,10,000

Total parts = 6 + 5 = 11 parts

Raj's share =  $(5/11) \times 1,10,000 = ₹50,000$  ?

Final Answer: Raj gets ₹50,000.

### Source: Partnership\Partnership-5

Problem on Partnership-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

M invests ₹4,000.

After 3 months, N joins with ₹8,000 and P joins with ₹20,000.

At the end of one year, the profit-sharing ratio is given as 6 : 9 : 5,

The total profit is ₹1,12,000.

Find the time of investment of P.

Solution:

Note: When the time of investment is different,

we multiply each partner's Investment by the time (in months) for which it was invested.

Rest the logic is the same ? the ratio of (Investment  $\times$  Time) will be the ratio in which the profit is divided.

M's investment = ₹4,000 for 12 months  $= 4,000 \times 12 = 48,000$ .

N's investment = ₹8,000 for  $(12 - 3) = 9$  months  $= 8,000 \times 9 = 72,000$ .

P's investment = ₹20,000 for  $t$  months  $= 20,000 \times t$ .

Given ratio = 6 : 9 : 5, so

$48,000 : 72,000 : (20,000 \times t) = 6 : 9 : 5$ .

Since M and N already match the ratio 6 : 9, we only need to compare N and P to find P's time.

So, compare N and P:

$(72,000) / (20,000 \times t) = 9 / 5$

Cross multiplying:

$72,000 \times 5 = 9 \times 20,000t$

$3,60,000 = 1,80,000t$

$t = 2$

Therefore  $t = 2$  months.?

Final Answer: P invested for 2 months.

**Source: Partnership\Partnership-6**

Problem on Partnership-6 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

A and V invest in the ratio 5 : 2.

After donating 10% of the profit to an orphanage, A gets ₹3,420 as profit.

Find the total profit.

Solution:

Note: The profit is always divided in the same ratio as the investment ratio.

Profit among the partners is distributed only after donating to an orphanage.

Since Investment ratio = Profit ratio = 5 : 2

Let total profit =  $100x$

Donation = 10% of  $100x = 10x$

Remaining profit =  $100x - 10x = 90x$

This  $90x$  is divided in the ratio 5 : 2 because after donating, the remaining profit will be distributed.

So, A's share =  $(5/7) \times 90x = (450/7)x$

Given A's share = 3,420

So,  $(450/7)x = 3,420$

$x = 53.2$

Total profit =  $100x = 100 \times 53.2 = 5,320$

Final Answer: The total profit = ₹5,320.

**Source: Partnership\Partnership-7**

Problem on Partnership-7 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

P and Q start a business by investing ₹20,000 and ₹40,000 respectively.

At the end of one year, the total profit is ₹15,000.

P also gets a fixed salary along with the profit share.

If P's total earning is ₹7,000, find the salary.

Solution:

Note: The profit is always divided in the same ratio as the investment ratio.

Salary will be distributed initially and after that profit is distributed in the profit sharing ratio.

P invests ₹20,000 and Q invests ₹40,000.

Investment ratio =  $20000 : 40000 = 1 : 2$ .

Total earnings includes salary and profit share.

Let P's salary =  $x$

Then profit left for division =  $15,000 - x$

P's share from profit =  $(1/3) \times (15,000 - x)$

Now, P's total earning = Salary + Profit share  
 $= x + (1/3)(15,000 - x)$

Given P's total earning = 7,000

$$x + (15,000 - x)/3 = 7,000$$

$$(3x + 15,000 - x)/3 = 7,000$$

$$x = 3,000$$

Final Answer: P's salary = ₹3,000.

### Source: Partnership\Partnership-8

Problem on Partnership-8 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of Partnership.

Problem:

Ram and Shyam together invested ₹15,000 in a business.

At the end of the year, the total profit was ₹2,000.

Shyam got ₹600 as his share of profit.

How much did Ram invest?

Solution:

Note: The profit is always divided in the same ratio as the investment ratio.

Given Total profit = ₹2,000

Shyam's share = ₹600

Ram's share =  $2,000 - 600 = ₹1,400$

So, profit ratio = Ram : Shyam =  $1,400 : 600 = 14 : 6 = 7 : 3$

Let Ram's investment =  $x$

Then Shyam's investment =  $(15,000 - x)$

Since profit is divided in the ratio of their investments,

so,  $(x / 15000 - x) = 7 / 3$

$$3x = 7(15,000 - x)$$

$$3x = 1,05,000 - 7x$$

$$x = 10,500$$

Final Answer: Ram invests ₹10,500.

**Source: Percentage\Percentage-1**

Problem on Percentage-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of percentage and some basic calculation.

Problem:

What percent is 42 kg of 336 kg?

Solution:

$$\text{Percentage} = (\text{Part} \div \text{Whole}) \times 100$$

$$= (42 \div 336) \times 100$$

$$= (1 \div 8) \times 100$$

$$= 12.5 \%$$

Final Answer:

42 kg is 12.5% of 336 kg.

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**Source: Percentage\Percentage-2**

Problem on Percentage-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of percentage and some basic calculation.

Problem:

If 15% of y is the same as 21% of z,  
then 12.5% of y is what percent of z?

Solution:

$$(15/100) y = (21/100) z$$

$$y = (21/15) z$$

$$y = (7/5) z$$

Now,

Let 12.5% of y = A % of z

$$12.5\% \text{ of } y = A \% \text{ of } z$$

Substitute  $y = (7/5) z$

12.5 % of y can be written as  $1/8$  of y after simplification

$$(1/8) \times (7/5) z = A \% \text{ of } z$$

$$A = (7/40) \times 100$$

$$A = 17.5 \%$$

Final Answer:

12.5% of y is 17.5% of z.

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**Source: Percentage\Percentage-3**

Problem on Percentage-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of percentage and some basic calculation.

Problem:

If the price of rice is 30% less than the price of wheat,  
then the price of wheat is how much percent more than the price of rice?

Solution:

Assume price of wheat = 100

Price of rice = 100 - 30% of 100

$$= 100 - 30$$

$$= 70$$

Difference between wheat and rice = 100 - 70 = 30

Percentage more = (Difference ÷ Price of rice) × 100

$$= (30 \div 70) \times 100$$

$$= 42.86\%$$

Final Answer:

Price of wheat is 42.86% more than the price of rice

**Source: Percentage\Percentage-4**

Problem on Percentage-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of percentage and some basic calculation.

Problem:

The price of apple is first increased by 10% and  
then decreased by 10%. What is the change in the price of apple?

Solution:

Note: Percentage change = (Difference ÷ Original price) × 100

Assume price of apple = 100

Increase by 10%

Price after increase = 100 + 10% of 100

$$= 100 + 10$$

$$= 110$$

Decrease by 10%

Price after decrease = 110 - 10% of 110

$$= 110 - 11$$

$$= 99$$

Difference = Original price - Final price

$$= 100 - 99$$

$$= 1$$

Percentage change = (Difference ÷ Original price) × 100

$$= (1 \div 100) \times 100$$

$$= 1\%$$

? Final Answer:

Price of apple decreases by 1%

### Source: Percentage\Percentage-5

Problem on Percentage-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of percentage and some basic calculation.

Problem:

An employee's salary is increased by 12% and then decreased by 8%.

If the original salary is ₹100, find the net percentage change.

Solution:

Note: Net percentage change = ( New value - Old value / Old value ) x 100

Consider Original salary = 100

Increase by 12%

12% of 100 = 12

New salary = 100 + 12 = 112

Decrease by 8%

8% of 112 = 8.96

New salary = 112 - 8.96 = 103.04

Compare with original

Original = 100,

Final = 103.04

Change = 103.04 - 100 = 3.04

$$\text{Net percentage change is} = (\text{Change} / \text{Original Salary}) \times 100$$

$$= (3.04 / 100) \times 100$$

$$= 3.04\%$$

?Final Answer:

Net percentage change is 3.04 %

### Source: Percentage\Percentage-6

Problem on Percentage-6 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of percentage and some basic calculation.

Problem:

Y has to score 40% marks to pass.  
He gets 20 marks and fails by 40 marks.  
The maximum marks of the exam are?

Solution:

Assume maximum marks =  $100x$

Marks required to pass = 40% of  $100x$   
 $= 40/100 \times 100x$   
 $= 40x$

Y gets 20 marks and fails by 40 marks, so passing marks = Y's marks + 40  
 $= 20 + 40$   
 $= 60$

Find actual maximum marks

If  $40x = 60$ , then

$$x = 60 \div 40$$

$$x = 1.5$$

Maximum marks =  $100x = 100 \times 1.5 = 150$  ?

? Final Answer:

Maximum marks of the exam = 150

### Source: Percentage\Percentage-7

Problem on Percentage-7 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.)  
because it makes the use of percentage and some basic calculation.

Problem:

In an election which was contested by 2 candidates,  
one candidate got 40% of total votes and yet lost by 1000 votes.  
Find the total number of votes cast in the election.

Solution:

Let total votes =  $100x$

Votes received by the losing candidate = 40% of total votes  
 $= 40\% \text{ of } 100x$   
 $= 40x$

Votes received by the winning candidate = Remaining votes =  $100x - 40x = 60x$

Difference in votes =  $60x - 40x = 20x$

This difference = 1000 votes

Solve for x  
 $20x = 1000$   
 $x = 50$

Total votes =  $100x = 100 \times 50 = 5000$  ?

? Final Answer:

Total number of votes cast in the election = 5000

### Source: Probability\Probability-1

Problem on Probability-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of probability logics.

Problem:

Two coins are tossed at the same time.

What is the probability of getting

- (a) two heads,
- (b) at least one head?

Solution:

Note: Probability = Total favourable outcomes/ Total outcomes

When 2 coins are tossed

Outcomes = {HH, HT, TH, TT}

So, total outcomes = 4

(a) Two heads:

Favourable outcomes = {HH} = 1

Probability = Total favourable outcomes/ Total outcomes

Probability =  $1 \div 4 = 1/4$  ?

(b) At least one head:

Atleast one head means 1 or more heads so,

Favourable outcomes = {HH, HT, TH} = 3

Probability = Total favourable outcomes/ Total outcomes

Probability =  $3 \div 4 = 3/4$  ?

Final Answer:

? (a) Probability of two heads =  $1/4$

? (b) Probability of at least one head =  $3/4$

### Source: Probability\Probability-2

Problem on Probability-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.)



because it makes the use of probability logics.

Problem:

A bag contains 5 red balls and 3 green balls.

One ball is picked at random.

What is the probability that the ball is red?

Solution:

Note: Probability = Favourable outcomes ÷ Total outcomes

Total balls =  $5 + 3 = 8$

Number of favourable outcomes (red balls) = 5

Probability = Favourable outcomes ÷ Total outcomes  
 $= 5 \div 8$  ?

Final Answer:

? Probability of picking a red ball =  $5/8$

### Source: Probability\Probability-3

Problem on Probability-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.)

because it makes the use of probability logics.

Problem:

A dice is thrown once.

What is the probability of getting

(a) an even number,

(b) a number greater than 4?

Solution:

Note: Probability = Favourable outcomes ÷ Total outcomes

Total outcomes when a dice is thrown = 6

(Possible numbers: 1, 2, 3, 4, 5, 6)

(a) Even number:

Even numbers = {2, 4, 6} ? 3 outcomes

Probability = Favourable outcomes ÷ Total outcomes

Probability =  $3 \div 6 = 1/2$  ?

(b) Number greater than 4:

Numbers greater than 4 = {5, 6} ? 2 outcomes

Probability = Favourable outcomes ÷ Total outcomes

Probability =  $2 \div 6 = 1/3$  ?

Final Answer:

? (a) Probability of even number =  $1/2$

? (b) Probability of number greater than 4 =  $1/3$

**Source: Probability\Probability-4**

Problem on Probability-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of probability logics.

Problem:

At a traffic signal, the light can be Red, Yellow, or Green with equal chance.

What is the probability that the light will be

- (a) Red,
- (b) Not Green?

Solution:

Note : Probability = Total favourable outcomes/ Total outcomes

Total possible outcomes = 3 (Red, Yellow, Green)

(a) Red

Favourable = 1 (Red)

Probability = Total favourable outcomes/ Total outcomes

Probability =  $1 \div 3 = 1/3$  ?

(b) Not Green

Favourable = {Red, Yellow} = 2

Probability = Total favourable outcomes/ Total outcomes

Probability =  $2 \div 3 = 2/3$  ?

Final Answer:

? (a) Probability of Red =  $1/3$

? (b) Probability of Not Green =  $2/3$

**Source: Probability\Probability-5**

Problem on Probability-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of probability logics.

Problem:

Three fair coins are tossed simultaneously.

Find the probability of getting atmost two heads.

Solution:

Note: Probability = Total favourable outcomes/ Total outcomes

Each coin can be Head (H) or Tail (T).

Sample Space  $S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$

?At most two heads? means 0, 1, or 2 heads.

Total favourable outcomes = 8 { HHT, HTH, THH, HTT, THT, TTH, TTT }

Probability = Total favourable outcomes/ Total outcomes

Probability =  $7/8$  ?

?Final Answer: Probability of at most two heads =  $7/8$ .

### Source: Probability\Probability-6

Problem on Probability-6 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of probability logics.

Problem:

Two dice are thrown together.

Find the probability that

- (a) both dice show the same number,
- (b) the sum is 8.

Solution:

Note: Probability = Total favourable outcomes/ Total outcomes

When 2 dice are thrown, each die has 6 faces.

So total outcomes =  $6 \times 6 = 36$ .

Sample space =

{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6),  
(2,1), (2,2), (2,3), (2,4), (2,5), (2,6),  
(3,1), (3,2), (3,3), (3,4), (3,5), (3,6),  
(4,1), (4,2), (4,3), (4,4), (4,5), (4,6),  
(5,1), (5,2), (5,3), (5,4), (5,5), (5,6),  
(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}

Total = 36 outcomes

(a) Both dice same number

Favourable outcomes = {(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)} = 6

Probability = Total favourable outcomes/ Total outcomes

Probability =  $6 \div 36 = 1/6$  ?

(b) Sum = 8

Favourable outcomes = {(2,6), (3,5), (4,4), (5,3), (6,2)} = 5

Probability = Total favourable outcomes/ Total outcomes

Probability =  $5 \div 36 = 5/36$  ?

Final Answer:

? (a) Probability of same number =  $1/6$

? (b) Probability sum = 8 ?  $5/36$

### Source: Probability\Probability-7

Problem on Probability-7 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of probability logics.

Problem:

A card is drawn at random from a pack of 52 cards.

What is the probability that it is

- (a) a red card,
- (b) a king,
- (c) face card (Jack, Queen, King)
- (d) Ace?

Solution:

Note: Probability = Total favourable outcomes/ Total outcomes

Total cards = 52

(a) Red cards in deck of cards = 26

Probability =  $26 \div 52 = 1/2$  ?

(b) Kings in deck of cards = 4

Probability =  $4 \div 52 = 1/13$  ?

(c) 4 suits: ? Hearts, ? Diamonds, ? Clubs, ? Spades

Each suit has 13 cards: A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K

The J (Jack), Q (Queen), and K (King) are called face cards because they have faces drawn on them.

Each suit has 3 face cards (J, Q, K)

Total =  $3 \times 4 = 12$  face cards in the whole pack

Face cards = 3 in each suit  $\times$  4 suits = 12

Probability =  $12 \div 52 = 3/13$  ?

(d) The A (Ace) is the first card of every suit.

There are 4 suits then there would be 4 aces in a deck of card.

Aces = 4

Probability =  $4 \div 52 = 1/13$  ?

?Final Answer:

Probability of Red cards =  $1/2$

Probability of Kings =  $1/13$

Probability of Face cards =  $3/13$

Probability of Ace =  $1/13$

### Source: Problem on Ages\Problem on Ages-1

Problem on Ages-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of basic linear equations.

Problem:

The present ages of A and B are in the ratio 3 : 5.  
 After 6 years, the ratio of their ages will be 2 : 3.  
 Find the present ages of A and B.

Solution:

Let the present age of A be =  $3x$

Let the present age of B be =  $5x$

After 6 years, A's age will be  $3x + 6$  and B's age will be  $5x + 6$ .

Given ratio after 6 years is 2 : 3

So,

$$(3x + 6) / (5x + 6) = 2 / 3.$$

$$3 \times (3x + 6) = 2 \times (5x + 6).$$

So,  $x = 6$ .

Put the value of  $x$  back:

$$A = 3 \times 6 = 18 \text{ years.}$$

$$B = 5 \times 6 = 30 \text{ years.}$$

Final Answer: A is 18 years old and B is 30 years old.

### Source: Problem on Ages\Problem on Ages-2

Problem on Ages-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of basic linear equations.

Problem:

Ram is three times as old as Sam.

Two years ago, he was five times as old as Sam.

What is the present age of Ram?

Solution:

Let the present age of Sam =  $x$

So, the present age of Ram =  $3x$

Two years ago:

$$\text{Sam's age} = x - 2$$

$$\text{Ram's age} = 3x - 2$$

According to the condition:

$$3x - 2 = 5(x - 2)$$

$$3x - 2 = 5x - 10$$

$$-2x = -8$$

$$x = 4$$

Now put the value of  $x$  back:

$$\text{Ram's present age} = 3 \times 4 = 12 \text{ years}$$

Final Answer:

? Ram's present age = 12 years.

### Source: Problem on Ages\Problem on Ages-3

Problem on Ages-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of basic linear equations.

Problem:

12 years ago, the age of P was 3 times the age of Q.

After 12 years, the ratio of ages of Q to P will be 2 : 3.

What is the present age of P?

Solution:

Let the present age of Q = x

So, the present age of P = y

12 years ago:

P's age = y - 12

Q's age = x - 12

According to the condition-1 ( 12 years ago ):

$$y - 12 = 3(x - 12)$$

$$y = 3x - 24 \quad \text{---(i)}$$

According to the condition-2 ( after 12 years ):

$$(x + 12) / (y + 12) = 2 / 3$$

$$3(x + 12) = 2(y + 12)$$

$$3x + 12 = 2y \quad \text{---(ii)}$$

Substitute (i) into (ii):

$$3x + 12 = 2(3x - 24)$$

$$3x + 12 = 6x - 48$$

$$60 = 3x$$

$$x = 20$$

Now from (i):

$$y = 3x - 24 = 3(20) - 24 = 60 - 24 = 36$$

So, P's present age = 36 years ?

Final Answer:

? P's present age = 36 years.

### Source: Problem on Ages\Problem on Ages-4

Problem on Ages-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of basic linear equations.

Problem:

The average age of Ram and Shyam is 65 years.

The average age of Ram, Shyam, and John is 53 years.

What is the age of John?

Solution:

Average age of Ram and Shyam = 65

Sum of observation = Average X No of observations

So, sum of their ages =  $65 \times 2 = 130$

Average age of Ram, Shyam, and John = 53

So, sum of their ages =  $53 \times 3 = 159$

Now,

John's age = (Sum of 3 persons) - (Sum of Ram and Shyam)

$= 159 - 130$

$= 29$

Final Answer:

John's age = 29 years

### Source: Problem on Ages\Problem on Ages-5

Problem on Ages-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of basic linear equations.

Problem:

The average age of 10 students and their teacher is 15 years.

The average age of the first seven students is 15 years and the average age of the last three students is 11 years.

What is the teacher's age?

Solution:

Note : Average = Sum of observation / No. of observation.

Average of 10 students + teacher = 15

Total people = 10 students + 1 teacher = 11

Total sum of age of all (10 students + teacher) =  $11 \times 15 = 165$

Average of first seven students = 15

Sum of first seven students =  $7 \times 15 = 105$

Average of last three students = 11

Sum of last three students =  $3 \times 11 = 33$

Total sum of age of all 10 students =  $105 + 33 = 138$

$$\begin{aligned}
 \text{Teacher's age} &= \text{Total age of all (11 persons)} - \text{Total age of 10 students} \\
 &= 165 - 138 \\
 &= 27
 \end{aligned}$$

Final Answer:

? Teacher's age = 27 years.

### Source: Problem on Ages\Problem on Ages-6

Problem on Ages-6 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of basic linear equations.

Problem:

The average age of a group of 4 friends is 36 years.

The youngest friend among them is 6 years old.

What was the average age of the group at the time of the birth of the youngest?

Solution:

Average = Sum of observation x No of observation

Average age of 4 friends = 36

So, total sum of age of 4 friends =  $36 \times 4 = 144$

Youngest friend's age = 6 years

? That means 6 years ago, this friend was born.

6 years ago:

Subtract 6 years from each of the 4 friends' ages

Total sum of age of group =  $144 - (6 \times 4)$

$$= 144 - 24 = 120$$

Now, we have to find the average at the time of birth of youngest

So, number of persons in the group will be = 3

Average age =  $120 \div 3 = 40$  ?

Final Answer:

? Average age of group 6 years ago =  $120 \div 3 = 40$  .

### Source: Problems on Train\Problem on Train-1

Problem on Train-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Accenture, TCS, Wipro, etc.) because it makes the use of distance, speed and time.

Problem:

A train runs at 54 km/h and passes a pole in 12 seconds.

Find the length of the train in meters.



Solution:

Note: Formula to be used to solve the problem:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

(Distance):

When a train crosses a pole, the distance covered = length of the train

(Speed in m/sec):

We know that to change km/h into m/s, we multiply by 5/18.

So,  $54 \times (5/18) = 15 \text{ m/s}$ .

(Time)

Given time to pass the pole = 12 sec.

Thus,

(Distance = Speed  $\times$  Time)

$$15 \times 12 = 180 \text{ meter.}$$

Final Answer:

The train is 180 meters long.

### Source: Problems on Train\Problem on Train-2

Problem on Train-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Accenture, TCS, Wipro, etc.) because it makes the use of distance, speed and time.

Problem:

A train 200 metres long runs at 60 km/hr.

Find the time taken to cross a bridge 300 metres long.

Solution:

Note: Formula to be used to solve the problem:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Therefore, Time = Distance / Speed

Now,

(Total distance):

$$\text{Distance} = \text{length of train} + \text{length of bridge} = 200 + 300 = 500 \text{ metres}$$

(Speed in m/sec):

$$60 \times (1000/3600) = 60 \times (5/18) = 16.67 \text{ m/sec}$$

Thus,

(Time = Distance  $\div$  Speed):

$$500 \div 16.67 \approx 30 \text{ seconds}$$

Final Answer:

?Thus train takes 30 seconds to cross.

### Source: Problems on Train\Problem on Train-3

Problem on Train-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Accenture, TCS, Wipro, etc.) because it makes the use of distance, speed and time.

Problem:

A train of length 180 meters passes a pole in 12 seconds.

After that, it passes a bridge in 30 seconds.

Find the length of the bridge in meters.

Solution:

Note: Formula to be used to solve the problem:

Distance = Speed  $\times$  Time

(Distance)

When a train crosses a pole, the distance covered = length of the train.

Distance = 180 meter.

Here, Time to cross a pole = 12 seconds

(Speed)

So, Speed = Distance  $\div$  Time =  $180 \div 12 = 15$  m/s.

(Time to cross the bridge)

When a train crosses a bridge, the distance covered = length of train + length of bridge.

Here, Time = 30 seconds and Speed = 15 m/s.

(Distance covered while crossing the bridge)

Distance = Speed  $\times$  Time =  $15 \times 30 = 450$  meters.

(Length of the bridge)

$450 = 180 + \text{Length of bridge}$

Length of bridge =  $450 - 180 = 270$  meters ?

Final Answer:

?The length of the bridge is 270 meters.

### Source: Problems on Train\Problem on Train-4

Problem on Train-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Accenture, TCS, Wipro, etc.) because it makes the use of distance, speed and time.

Problem:

A train passes a pole in 10 seconds and

the same train passes a platform of length 200 meters in 30 seconds.  
Find the speed of the train in m/s and the length of the train meters.

Solution:

Note: Formula to be used to solve the problem:

Distance = Speed  $\times$  Time

(Train crossing the pole)

When a train crosses a pole, the distance covered = length of the train.

So, Length of train = Speed  $\times$  10

(Train crossing the platform)

When a train crosses a platform, the distance covered = length of train + length of platform.

So, Length of train + 200 = Speed  $\times$  30

(Equation formation)

From pole: Length of train = 10  $\times$  Speed

From platform: Length of train + 200 = 30  $\times$  Speed

(Substitute)

10  $\times$  Speed + 200 = 30  $\times$  Speed

200 = 20  $\times$  Speed

Speed = 10 m/s

(Length of train)

Length of train = 10  $\times$  Speed = 10  $\times$  10 = 100 meters ?

Final Answer:

?The speed of the train is 10 m/s and the length of the train is 100 meters.

### Source: Problems on Train\Problem on Train-5

Problem on Train-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Accenture, TCS, Wipro, etc.) because it makes the use of distance, speed and time.

Problem:

Two trains are moving towards each other on parallel tracks.

Train A is 150 meters long and runs at 54 km/h.

Train B is 120 meters long and runs at 36 km/h.

Find the time taken for the two trains to completely cross each other in seconds.

Solution:

Note: Formula to be used:

Distance = Speed  $\times$  Time

Concept of relative speed:

If two objects move towards each other, their relative speed = sum of their speeds.

If they move in the same direction, their relative speed = difference of their speeds.

(Relative speed in m/s)

To convert km/h to m/s, multiply by  $5/18$ .

Speed of Train A =  $54 \times (5/18) = 15$  m/s.

Speed of Train B =  $36 \times (5/18) = 10$  m/s.

Relative speed =  $15 + 10 = 25$  m/s (since trains move towards each other).

(Distance to be covered)

When two trains cross each other completely, distance covered = sum of their lengths =  $150 + 120 = 270$  meters.

(Time to cross)

Time = Distance  $\div$  Relative speed =  $270 \div 25 = 10.8$  seconds.?

Final Answer:

?The two trains will completely cross each other in 10.8 seconds.

### Source: Problems on Train\Problem on Train-6

Problem on Train-6 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Accenture, TCS, Wipro, etc.) because it makes the use of distance, speed and time.

Problem:

A train 180 meters long is running at 54 km/h.

A man is walking on the track in the same direction at 6 km/h.

Find the time taken by the train to completely cross the man in seconds.

Solution:

Note: Formula to be used:

Distance = Speed  $\times$  Time

Concept of relative speed:

If two objects move towards each other, relative speed = sum of their speeds.

If they move in the same direction, relative speed = difference of their speeds.

(Relative speed in m/s)

Convert both speeds into m/s by multiplying with  $5/18$ .

Train's speed =  $54 \times (5/18) = 15$  m/s.

Man's speed =  $6 \times (5/18) = 5/3 \approx 1.67$  m/s.

Relative speed =  $15 - 1.67 = 13.33$  m/s.

(Distance to be covered)

The train has to cover its own length to cross the man = 180 meters.

(Time to cross)

Time = Distance  $\div$  Relative speed =  $180 \div 13.33 \approx 13.5$  seconds.?

Final Answer

?The train will completely cross the man in 13.5 seconds.

**Source: Problems on Train\Problem on Train-7**

Problem on Train-7 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Accenture, TCS, Wipro, etc.) because it makes the use of distance, speed and time.

Problem:

Stations M and N are 276 km apart.

Train A starts from M towards N at 30 km/h at the same time Train B starts from N towards M at 16 km/h.

Find after how many hours they meet?

Solution:

Note: Formula to be used:

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Concept of relative speed:

If two objects move towards each other, their relative speed = sum of their speeds.

If they move in the same direction, their relative speed = difference of their speeds.

So, Relative speed =  $30 + 16 = 46$  km/h.

Time until meeting = Distance  $\div$  Relative speed =  $276 \div 46 = 6$  hours ?

Final Answers:

? They meet after 6 hours.

**Source: Profit & Loss\Profit and Loss-1**

Problem on Profit & Loss-1 ? Placement Exercise

NOTE: This kind of question is very common in campus placements (TCS, Infosys, Wipro, Capgemini, etc.) because it mixes marked price, discount, and selling price.

Problem

Ajay incurred a loss of 20% by selling a vase for Rs. 2880.

To get a profit of 20%, at what price should he sell the vase?

Solution

Let cost price (CP) =  $100x$

Loss is calculated on Cost Price

Loss = 20% on CP

$$= 20x$$

Selling price at 20% loss = CP - Loss

$$= 100x - 20x$$

$$= 80x$$

According to ques, SP = 2880

$$80x = 2880$$

$$x = 2880 \div 80$$

$$x = 36$$

$$\text{Cost price} = 100x = 100 \times 36 = 3600 ?$$

$$\begin{aligned}\text{Profit} &= 20\% \text{ of CP} \\ &= 20\% \text{ of } 3600\end{aligned}$$

$$\begin{aligned}\text{Selling price for } 20\% \text{ profit} &= \text{Cost price} + \text{Profit} \\ &= 3600 + 720 \\ &= 4320\end{aligned}$$

? Final Answer:

To get a profit of 20%, Ajay should sell the vase for Rs. 4320

### Source: Profit & Loss\Profit and Loss-2

Problem on Profit & Loss-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of profit and loss.

Problem:

A shopkeeper buys a shirt for ₹800.  
He marks its price 25% above the cost price.  
He then gives a 10% discount on the marked price and sells it.  
Find profit percentage.

Solution:

$$\text{Note: Profit \%} = (\text{Profit} / \text{CP}) \times 100$$

Find Marked Price (MP)

$$\begin{aligned}\text{MP} &= \text{CP} + 25\% \text{ of CP} \\ \text{MP} &= 800 + 25\% \times 800 \\ \text{MP} &= 1000\end{aligned}$$

Apply discount of 10% on MP

$$\begin{aligned}\text{Discount} &= 10\% \text{ of } 1000 \\ \text{Discount} &= 100\end{aligned}$$

$$\text{SP} = \text{MP} - \text{Discount}$$

$$\text{SP} = 1000 - 100 = 900$$

$$\text{Profit} = \text{SP} - \text{CP}$$

$$\text{Profit} = 900 - 800 = 100$$

$$\text{Profit \%} = (\text{Profit} / \text{CP}) \times 100$$

$$\text{Profit \%} = (100/800) \times 100 = 12.5\%$$

?Final Answer: The Profit Percentage will be 12.5%

### Source: Profit & Loss\Profit and Loss-3

**Problem on Profit & Loss-3 ? Placement Exercise**

NOTE: This kind of question is very common in campus placements (TCS, Infosys, Wipro, Capgemini, etc.) because it mixes marked price, discount, and selling price.

**Problem**

A shopkeeper buys an article at a discount of 20% on the marked price.  
He then sells it at a price which is 20% more than the marked price.  
Find his profit percentage on the cost price.

**Solution:**

Note: Profit % =  $(\text{Profit} \div \text{CP}) \times 100$

Assume Marked Price (MP) = ₹100

Cost Price (CP):

Discount = 20% of 100 = 20

CP =  $100 - 20 = ₹80$

Selling Price (SP):

SP = MP + 20% of MP

=  $100 + 20 = ₹120$

**Profit:**

Profit = SP - CP =  $120 - 80 = ₹40$

**Profit %:**

Profit % =  $(\text{Profit} \div \text{CP}) \times 100 = (40 \div 80) \times 100 = 50\%$

**Final Answer**

∴ The shopkeeper makes a 50% profit.

**Source: Profit & Loss\Profit and Loss-4****Problem on Profit & Loss-4 ? Placement Exercise**

NOTE: This kind of question is very common in campus placements (TCS, Infosys, Wipro, Capgemini, etc.) because it mixes marked price, discount, and selling price.

**Problem:**

A sold a car to B at a profit of 25%.

B incurred a loss of 15% while selling the same car to C.

A spent Rs. 50000 for this car.

Find the price at which C bought it.

**Solution:**

Cost price for A = 50000

Selling price of A to B = CP + 25% of 50000

=  $50000 + 25\% \text{ of } 50000$

=  $50000 + 12500$

$$= 62500$$

B sold to C at a loss of 15%,

So selling price of B to C = CP ? 15% of 62500

$$= 62500 - 15\% \text{ of } 62500$$

$$= 62500 - 9375$$

$$= 53125$$

? Final Answer:

C bought the car for Rs. 53125

### Source: Profit & Loss\Profit and Loss-5

Problem on Profit & Loss-5 ? Placement Exercise

NOTE: This kind of question is very common in campus placements (TCS, Infosys, Wipro, Capgemini, etc.) because it mixes marked price, discount, and selling price.

Problem

A fruit seller buys lemons at 2 for a rupee and sells them at 5 for 3 rupees.

Find his gain percent.

Solution

Note: Gain % = (Gain / Cost price) X 100

Cost price of 2 lemons = 1 rupee

Cost price of 1 lemon =  $1 \div 2 = 0.5$  rupee

Selling price of 5 lemons = 3 rupees

Selling price of 1 lemon =  $3 \div 5 = 0.6$  rupee

Gain on 1 lemon = Selling price - Cost price

$$= 0.6 - 0.5$$

$$= 0.1 \text{ rupee}$$

Gain percent = (Gain ÷ Cost price) × 100

$$= (0.1 \div 0.5) \times 100$$

$$= 20\%$$

? Final Answer:

Gain percent = 20%

### Source: Profit & Loss\Profit and Loss-6

Problem on Profit & Loss-6 ? Placement Exercise

NOTE: This kind of question is very common in campus placements (TCS, Infosys, Wipro, Capgemini, etc.) because it mixes marked price, discount, and selling price.

Problem



A shopkeeper earns a profit of 15% after selling a book at 20% discount on the printed price.  
Find the ratio of the cost price and printed price of the book.

Solution:

Let the printed price =  $100x$

Selling price after 20% discount =  $100x - 20\% \text{ of } 100x$   
 $= 100x - 20x$   
 $= 80x$

Let the cost price =  $100y$

Profit = 15% of cost price  
 $= 15\% \text{ of } 100y$

Selling price = Cost price + Profit

$80x = 100y + 15\% \text{ of } 100y$

$80x = 100y + 15y$

$80x = 115y$

$y : x = 80 : 115 = 16 : 23$

Final Answer:

The ratio of cost price to printed price =  $16 : 23$

### Source: Profit & Loss\Profit and Loss-7

Problem on Profit & Loss-7 ? Placement Exercise

NOTE: This kind of question is very common in campus placements (TCS, Infosys, Wipro, Capgemini, etc.) because it mixes marked price, discount, and selling price.

Problem

A cheater manipulates his weighing machine so that it shows 1 kg for 970 grams.  
Find his profit percent.

Solution

Note: In case of manipulation profit percentage can be written as-

Profit % =  $(\text{Actual weight} - \text{Given weight}) \div \text{Given weight} \times 100$

Let the cost price of 1 kg of goods = 100 (taking 1 kg as reference)

He actually gives 970 grams but charges for 1 kg.

Profit =  $(\text{Actual weight} - \text{Given weight}) \div \text{Given weight} \times 100$

Profit % =  $((1000 - 970) \div 970) \times 100$

Profit % =  $(30 \div 970) \times 100$

$= 3/97 \times 100$

$\approx 3.09\%$  (approx)

Final Answer:

Profit percent ? 3.09%

### Source: Ratio and Proportion\Ratio and Proportion-1

Ratio and Proportion-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.) because it makes the use of concepts of percentage and simplification along with the concepts of ratio.

Problem:

One number is 20% less than the third number,  
and the second number is 40% less than the third number.  
Find the ratio of the two numbers.

Solution:

(Assume third number = 100):

First number = 100 ? 20% of 100 = 80

Second number = 100 ? 40% of 100 = 60

Third number = 100

(Form the ratio):

First : Second = 80 : 60  
= 4 : 3 ?

Final Answer

? The ratio of the two numbers is 4 : 3

### Source: Ratio and Proportion\Ratio and Proportion-2

Ratio and Proportion-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.) because it makes the use of concepts of percentage and simplification along with the concepts of ratio.

Problem:

40% of one number is equal to  $\frac{2}{3}$  of another number.  
Find the ratio of the two numbers.

Solution:

Let the first number = A

the second number = B.

According to the question,

40% of A =  $\frac{2}{3}$  of B

$$A / B = 5 / 3$$

So, A : B = 5 : 3 ?

Final Answer

? The ratio of the two numbers is 5 : 3.

**Source: Ratio and Proportion\Ratio and Proportion-3**

Ratio and Proportion-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.)

because it makes the use of concepts of percentage and simplification along with the concepts of ratio.

Problem:

One number is 20% more than the third number,

The second number is 50% more than the third number.

Find the ratio of the three numbers.

Solution:

(Assume third number = 100):

First number =  $100 + 20\% \text{ of } 100 = 120$ Second number =  $100 + 50\% \text{ of } 100 = 150$ 

Third number = 100

(Form the ratio):

First : Second : Third = 120 : 150 : 100

Thus, 12 : 15 : 10?

Final Answer

?The ratio of three numbers is 12 : 15 : 10

**Source: Ratio and Proportion\Ratio and Proportion-4**

Ratio and Proportion-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.)

because it makes the use of concepts of percentage and simplification along with the concepts of ratio.

Problem:

If  $0.75 : x :: 5 : 8$ , then x is equal to:

Solution:

We know that,

 $a : b :: c : d$  means  $a/b = c/d$ 

So,

 $0.75 / x = 5 / 8$  $x = 6 \div 5$  $x = 1.2$  ?

Final Answer:

?x = 1.2

**Source: Ratio and Proportion\Ratio and Proportion-5**

## Ratio and Proportion-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.) because it makes the use of concepts of percentage and simplification along with the concepts of ratio.

Problem:

The ratio of two numbers is 2 : 3  
and their product is 726.  
Find the two numbers.

Solution:

Let the numbers be  $2x$  and  $3x$

$$\text{Product} = (2x) \times (3x) = 6x^2$$

(Given product = 726):

So according to the question:

$$6x^2 = 726$$

$$x = 11$$

$$\text{First number} = 2x = 2 \times 11 = 22$$

$$\text{Second number} = 3x = 3 \times 11 = 33 ?$$

Final Answer

? The two numbers are 22 and 33

**Source: Time and Work\Time and Work-1**

## Time and Work-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.) because it makes the use of concepts of work and efficiency.

Problem:

A can finish a piece of work in 15 days.  
B can finish the same work in 10 days.  
In how many days will they finish the same piece of work together?

Solution:

Note: Formula to be used to solve the problem

$$\text{Work} = \text{Efficiency} \times \text{Time}$$

(Take LCM of 15 and 10 for total work):

LCM(15, 10) = 30 units of work (assumed total work).

(Find efficiency of each):

$$A?s \text{ 1 day work} = 30 \div 15 = 2 \text{ units/day}$$

$$B?s \text{ 1 day work} = 30 \div 10 = 3 \text{ unit/day}$$

(Work done together in 1 day):

$$2 + 3 = 5 \text{ units/day}$$

(Total time = Total work ÷ Efficiency):

$$30 \div 5 = 6 \text{ days?}$$

Final Answer

?A and B will finish the work together in 6 days.

### Source: Time and Work\Time and Work-2

Time and Work-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.) because it makes the use of concepts of time, work and efficiency.

Problem:

A and B together can complete a piece of work in 10 days.

B and C together can complete it in 12 days.

C and A together can complete it in 15 days.

Find in how many days A, B, and C can complete same piece of work if they work together.

Solution:

Note: Formula to be used to solve the problem

$$\text{Work} = \text{Efficiency} \times \text{Time}$$

(Take LCM for total work)

LCM of 10, 12, and 15 = 60 units of total work.

(Find efficiency of each pair)

$$A + B \text{ efficiency} = 60 \div 10 = 6 \text{ units/day}$$

$$B + C \text{ efficiency} = 60 \div 12 = 5 \text{ units/day}$$

$$C + A \text{ efficiency} = 60 \div 15 = 4 \text{ units/day}$$

(Add all efficiencies)

$$(A + B) + (B + C) + (C + A) = 6 + 5 + 4 = 15$$

But this equals 2(A + B + C).

$$\text{So, } A + B + C \text{ efficiency} = 15 \div 2 = 7.5 \text{ units/day}$$

(Time taken together)

$$\text{Total Work} \div \text{Efficiency} = 60 \div 7.5 = 8 \text{ days ?}$$

Final Answer:

? A, B, and C will finish the work together in 8 days.

### Source: Time and Work\Time and Work-3

Time and Work-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.) because it makes the use of concepts of time, work and efficiency.

Problem:

Q can complete a work in 30 days.

Q is 25% more efficient than P.

Find the time taken by P alone to complete the same work.

Solution:

Note: Formula to be used to solve the problem

$$\text{Work} = \text{Efficiency} \times \text{Time}$$

(Assume efficiencies)

Let P's efficiency = 100.

Then Q's efficiency = 125 (since Q is 25% more efficient).

Ratio of Efficiency of P : Q = 4 : 5

(Find total work using Q)

$$\text{Work} = \text{Efficiency} \times \text{Time}$$

$$= 5 \times 30 = 150 \text{ units}$$

(Time taken by P)

$$\text{Time} = \text{Work} \div \text{Efficiency of P}$$

$$= 150 \div 4 = 37.5 \text{ days?}$$

Final Answer:

P alone can complete the work in 37.5 days.

### Source: Time and Work\Time and Work-4

Time and Work-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement (Infosys, TCS, Wipro, etc.)

because it makes the use of concepts of time, work and efficiency.

Problem:

4 men can complete a work in 12 days, and 6 boys can complete the same work in 12 days.

If 3 men and 9 boys work together, in how many days will they finish the work?

Solution:

Note: Formula to be used to solve the problem

$$\text{Work} = \text{Efficiency} \times \text{Time}$$

Work by 4 men in 12 days = Work by 6 boys in 12 days

So,

$$4M \times 12 = 6B \times 12$$

$$4M = 6B$$

$$B = \frac{2}{3} M$$

Now, 9 boys =  $9 \times (\frac{2}{3})M = 6 \text{ men}$ .

So, 3 men + 9 boys = 3 men + 6 men = 9 men.

So, We have written boys in form of men.

Consider that 9 men takes  $y$  days to complete the work.  
 We can equate the equations again that include the men,  
 Work by 4 men in 12 days = Work by 9 men in  $x$  days  
 So,

$$4M \times 12 = 9M \times y$$

$$y = 5? \text{ days ? (On cancelling M)}$$

Final Answer:

?They will finish the work in 5? days.

### Source: Time, Speed and Distance\Time, Speed and Distance-1

Problem on Time, Speed and Distance-1 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.)  
 because it makes the use of simple formula and calculations.

Problem:

A car is running at a speed of 72 km/h.  
 Find the time it will take to cover 6000 meters.

Solution:

Note: Distance = Speed X Time

Convert speed into m/s:

$$\text{Speed} = 72 \times (1000 \div 3600) = 20 \text{ m/s}$$

$$\text{Distance} = 6000 \text{ m}$$

$$\begin{aligned} \text{Time} &= \text{Distance} \div \text{Speed} \\ &= 6000 \div 20 \\ &= 300 \text{ seconds ?} \end{aligned}$$

? Final Answer:

The car will take 300 seconds to cover 6000 m.

### Source: Time, Speed and Distance\Time, Speed and Distance-2

Problem on Time, Speed and Distance-2 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.)  
 because it makes the use of simple formula and calculations.

Problem:

Ramesh sees a thief at a distance of 80 m.  
 The thief runs at 5 m/s, and Ramesh chases him at 7 m/s.  
 Find the distance the thief covers before Ramesh catches him.

Solution:

Note: Concept of relative speed:

If two objects move towards each other, relative speed = sum of their speeds.

If they move in the same direction, relative speed = difference of their speeds.

Relative speed of Ramesh with respect to the thief =  $7 - 5 = 2$  m/s

Time taken to catch the thief = Distance  $\div$  Relative speed

$$= 80 \div 2$$

$$= 40 \text{ seconds}$$

Distance covered by the thief = Speed of thief  $\times$  Time

$$= 5 \times 40$$

$$= 200 \text{ m ?}$$

? Final Answer:

The thief covers 200 m before Ramesh catches him.

### Source: Time, Speed and Distance\Time, Speed and Distance-3

Problem on Time, Speed and Distance-3 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of simple formula and calculations.

Problem:

A train travels 180 km at a speed of 60 km/h.

On the return journey, its speed is 90 km/h.

Find the average speed of the train for the whole journey.

Solution:

For finding average speed for equal distances there is a formula

$$\text{Avg Speed} = (2 \times S_1 \times S_2) / (S_1 + S_2)$$

According to the question

$$S_1 = 60 \text{ km/hr}$$

$$S_2 = 90 \text{ km/hr}$$

$$\text{Avg Speed} = (2 \times 60 \times 90) / (60 + 90) = 72 \text{ km/hr ?}$$

?Final Answer: The average speed is 72 km/h.

### Source: Time, Speed and Distance\Time, Speed and Distance-4

Problem on Time, Speed and Distance-4 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of simple formula and calculations.

Problem:



Rohit covers one-fourth of the total distance at 20 km/hr,  
one-fourth at 10 km/h and rest of his journey at 80 km/h.  
Find Rohit's average speed for the whole distance?

Solution:

Average speed is the total distance traveled divided by the total time taken.

Let total distance =  $100x$

Distance 1 =  $\frac{1}{4}$  of  $100x = 25x$ ,

Speed 1 = 20 km/h

Time 1 =  $25x \div 20 = 1.25x$  hours

Distance 2 =  $25x$ ,

Speed 2 = 10 km/h

Time 2 =  $25x \div 10 = 2.5x$  hours

Distance 3 =  $50x$ ,

Speed 3 = 80 km/h

Time 3 =  $50x \div 80 = 0.625x$  hours

Total time =  $1.25x + 2.5x + 0.625x = 4.375x$  hours ?

Average speed = Total distance  $\div$  Total time

$$= 100x \div 4.375x$$

$$= 100 \div 4.375$$

$$= 22.86 \text{ km/h ?}$$

? Final Answer:

Rohit's average speed ? 22.86 km/h

### Source: Time, Speed and Distance\Time, Speed and Distance-5

Problem on Time, Speed and Distance-5 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.)

because it makes the use of simple formula and calculations.

Problem:

Pratik travels 96 kms at a speed of 16 km/hr using a bike, 124 kms at 31 km/h by car and another 105 kms at 7 km/h in horse cart.

Then, find his average speed for the entire distance travelled?

Solution:

Average speed is the total distance traveled divided by the total time taken.

Distance 1 = 96 km,

Speed 1 = 16 km/h

Time 1 =  $96 \div 16 = 6$  hours

Distance 2 = 124 km,  
 Speed 2 = 31 km/h  
 Time 2 =  $124 \div 31 = 4$  hours

Distance 3 = 105 km,  
 Speed 3 = 7 km/h  
 Time 3 =  $105 \div 7 = 15$  hours

Total distance =  $96 + 124 + 105 = 325$  km  
 Total time =  $6 + 4 + 15 = 25$  hours

Average speed = Total distance  $\div$  Total time  
 $= 325 \div 25$   
 $= 13$  km/h ?

? Final Answer:  
 Pratik's average speed for the entire journey = 13 km/h

### Source: Time, Speed and Distance\Time, Speed and Distance-6

Problem on Time, Speed and Distance-6 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of simple formula and calculations.

Problem:

Surendra travels from home to office.

With an average speed of 50 km/h, he is late by 30 minutes.

With a speed of 60 km/h, he reaches 10 minutes early.

Find the distance between home and office.

Solution:

Note: Add times when one is late and one is early ? because one increases time, the other decreases it  
 Subtract times when both are late (or both early) ? because we want the difference due to speed change

Time = Distance  $\div$  Speed

Let the distance between home and office = D km

Time difference between the two speeds =  $30 + 10 = 40$  minutes =  $40/60$  hours =  $2/3$  hours ?

Time = Distance  $\div$  Speed

Time at 50 km/h ? Time at 60 km/h =  $2/3$  hours

$(D \div 50) - (D \div 60) = 2/3$

Take LCM of 50 and 60 = 300

$(6D - 5D) \div 300 = 2/3$

$D \div 300 = 2/3$

$D = 300 \times 2/3$

$D = 200$  km ?

? Final Answer:

Distance between home and office = 200 km

### Source: Time, Speed and Distance\Time, Speed and Distance-7

Problem on Time, Speed and Distance-7 - Placement Exercise

NOTE: This kind of question is very common in campus placement ( TCS, Wipro, etc.) because it makes the use of simple formula and calculations.

Problem:

Rohit drives from his home at a speed of 30 km/hr and reaches his bank 20 minutes late. Then the next day he increases his speed by 15 km/hr but still he is late by 8 minutes. How far is his bank from his home?

Solution:

Note: Add times when one is late and one is early ? because one increases time, the other decreases it  
 Subtract times when both are late (or both early) ? because we want the difference due to speed change  
 $\text{Time} = \text{Distance} \div \text{Speed}$

Let the distance between home and bank = D km

Time difference between the two speeds = 20 ? 8 = 12 minutes = 12/60 hours = 1/5 hours ?

Time at 30 km/h ? Time at 45 km/h = 1/5 hours

$(D \div 30) ? (D \div 45) = 1/5$

Take LCM of 30 and 45 = 90

$(3D ? 2D) \div 90 = 1/5$

$D \div 90 = 1/5$

$D = 18 \text{ km ?}$

? Final Answer:

Distance between Rohit's home and bank = 18 km

# Logical Reasoning

**Source: TATA ELXSI\Analytical.txt**

Syllogism:

In syllogism questions, you need to arrive at the proper conclusion based on the given premises.

Q1)

Statements:

Some mangoes are yellow.

Some apples are mangoes.

Conclusions:

(i) Some mangoes are green.

(ii) Apples is yellow.

a) Only conclusion (i) follows

b) Only conclusion (ii) follows

c) Either (i) or (ii) follows

d) Neither (i) nor (ii) follows <correct>

-----

Q2)

Statements:

All tigers are rats.

Some rats are lions.

Conclusions:

(i) Some tigers are lions.

(ii) Some lions are tigers.

a) Only conclusion (i) follows

b) Only conclusion (ii) follows

c) Either (i) or (ii) follows

d) Neither (i) nor (ii) follows <correct>

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Q3)

Statments:

All pencils are bricks.

All bricks are bottles.

Conclusions:

(i) All pencils are bottles.

(ii) All bricks are pencils.

a) Only conclusion (i) follows <correct>

b) Only conclusion (ii) follows

- c) Either (i) or (ii) follows
  - d) Neither (i) nor (ii) follows
- 

Q4)

Statements:

All hary are birds.  
All birds are crows.  
All crows are cats.

Conclusions:

- a) All birds are cats
  - b) All crows are hary
- 
- a) If only conclusion I is true.    <correct>
  - b) If only conclusion II is true.
  - c) If either conclusion I or II is true.
  - d) If neither conclusion I nor conclusion II is true.
- 

Q5)

Statements:

Some cows are crows.  
Some crows are elephants.

Conclusion:

- (i) Some cows are elephants.
  - (ii) All crows are elephants.
- 
- a) Only conclusion (i) follows    <correct>
  - b) Only conclusion (ii) follows
  - c) Either (i) or (ii) follows
  - d) Neither (i) nor (ii) follows

Directions for Questions 6-10:

Data sufficiency questions consist of a question followed by two statements. Need to decide whether the given information in the statements (taken singly or together) is sufficient to answer the question.

6) How many students passed the exam if 20 students failed in the exam?

I. Thousand students were issued hall tickets for the exam.

II. 10% of the students who appeared for the exam failed.

- a) If the question cannot be answered even with the help of both the statements together.
- b) If the question can be answered with the help of statement II alone.
- c) If the question can be answered with the help of statement I alone.
- d) If the question can be answered with the help of both statements together.

Ans: b

Explanation:

Information given in the question has to be noted first. The number of students who failed is given as 20. The statement I alone is not sufficient since the number of students who appeared is not known.

Using statement II alone, if 10% of the students appeared have failed, then 90% of students appeared have passed. It is given in the question that 20 students have failed. 10% of appeared = 20, 90% of appeared can be found.

7) What is the value of X?

I.  $X^4 = 2401$

II.  $X^2 + 3X - 24 = 0.5(6X + 50)$

- a) If the question cannot be answered even with the help of both the statements together.
- b) If the question can be answered with the help of statement II alone.
- c) If the question can be answered with the help of statement I alone.
- d) If the question can be answered with the help of both statements together.

Ans: a

Explanation:

From statement I, we obtain  $X = +7$  or  $-7$ . As we cannot uniquely determine the value of X, statement I alone is not sufficient to answer the question.

From statement II, we obtain  $X = +7$  or  $-7$ . As we cannot uniquely determine the value of X. So, even the two statements together are not sufficient to answer the question.

Hence the answer is a.

8) If a salesman received a commission of 3% of the sales that he has booked in a month, what was the sale booked by the salesman in the month of November 2003?

I. The sales booked by the salesman in the month of November 2003 minus salesman's commission was \$245,000

II. The selling price of the sales booked by the salesman in the month of November 2003 was 125 percent of the original purchase price of \$225,000

- a) If the question can be answered with the either of the statement alone.
- b) If the question can be answered with the help of statement II alone.
- c) If the question can be answered with the help of statement I alone.
- d) If the question can be answered with the help of both statements together.

Ans: a

Explanation:

From statement 1, we know the sales value after the salesman's commission is subtracted.

From the question stem, we know his commission is 3% of the sales booked. Then the value of sales after subtracting his commission is  $100 - 3 = 97\%$  of the sales booked. Putting the two together, we can deduce that 97% of sales booked = \$245,000. So we can find out the sales booked.

9) Is the positive integer  $m$  divisible by 6?

I.  $m$  is divisible by 3

II.  $m$  is divisible by 4

- a) If the question cannot be answered even with the help of both the statements together.
- b) If the question can be answered with the help of statement II alone.
- c) If the question can be answered with the help of statement I alone.
- d) If the question can be answered with the help of both statements together.

Ans: d

Explanation:

From statement 2, if  $m$  is divisible by 4, it will definitely be divisible by 2.

So, by combining the two statements, we know that  $m$  is divisible by 3 and by 2. Hence, we can conclude that  $m$  is divisible by 6.

10) What is the distance between Chandigarh and Delhi?

I. Karnal is 130 km from Chandigarh.

II. Delhi is 120 km from Karnal.

a) If the question cannot be answered even with the help of both the statements together.

b) If the question can be answered with the help of statement II alone.

c) If the question can be answered with the help of statement I alone.

d) If the question can be answered with the help of both statements together.

Ans: a

Explanation:

Chandigarh, Karnal, and Delhi are in a straight line and Karnal lies between Chandigarh and Delhi.

Even if it is given that these 3 cities are in straight line, still we have 2 possible answers to this question, even after combining the two statements i.e 250 km and 10 km.

Since we are not getting any unique answer even after combining the two statements.

Data Arrangements 11-12:

11) In a college there are seven clubs: Drama, Astronomy, Music, Dance, Cookery, Debate and Science. Each of these clubs meets on a Different day of the week, one on each day beginning Sunday. The Drama club must meet on a Sunday. The Astronomy club meeting is held after both the Music and Dance club meeting are held. The Cookery, Debate and Science meetings must be held on consecutive days, though not in the same order.

The music club meeting can be held, latest, on?



- a) Tuesday
- b) Wednesday
- c) Thursday
- d) Friday

Ans: d

Explanation:

Since the drama club must be on a Sunday and the Astronomy Club must meet after the music club meeting can be held, latest on Friday.

12) Which of the following clubs meet on 3 consecutive days?

- (I) Drama, Dance, Astronomy
- (II) Astronomy, Music, Debate
- (III) Dance, Cookery, Drama

- a) II only
- b) III only
- c) II and III only
- d) None

Ans: d

Explanation:

The Science, Debate and Cookery Clubs can meet on three consecutive days.

Data Arrangements 13-16:

In a community dance of Arunachal Pradesh, eight persons form a circular group. In Republic day parade a group from

Arunachal Pradesh consisting of Kavita, Laxmi, Munni, Nini, Oliver, Pallavi, Qunicy, Rehana presented this dance. All of them were related to one another due to which there were some constraints as to who would be beside whom. Laxmi, Munni, Nini -the only three women in the group -have to be side by side; Munni and Oliver have to be farthest from each other, and Qunicy and Kavita should have at least 3 others between them.

13) If Nini is placed opposite to Pallavi and between Qunicy and Munni, then which of the following should be opposite to each other?

- a) Munni and Qunicy
- b) Laxmi and Rehana
- c) Oliver and Pallavi
- d) Pallavi and Rehana

Ans: b

Explanation:

The arrangement will be

2- Munni, 3- Nini, 4- Quincy, 5- Rehana, 6- Oliver, 7- Pallavi, 8- Kavita, 1- Laxmi, so, Laxmi and Rehana are opposite to each other.

14) If Rehana is placed between Oliver and Qunicy, then Pallavi will be placed between?

- a) Oliver and Munni
- b) Oliver and Kavita
- c) Laxmi and Oliver
- d) Nini and Qunicy

Ans: b

Explanation:

6- Oliver

5- Rehana

4- Quincy

3

2- Munni

1

8- Kavita

7- Pallavi

Thus, Pallavi is placed between Oliver and Kavita.

15) If Laxmi is to be immediate left of Quincy, then Munni should be on the immediate right of?

a) Munni only

b) Quincy only

c) Kavita or Munni

d) Munni or Nini

Ans: c

Explanation:

1- L

2- (N)

3- (N)

4- Kavita

5-

6-

7-

8- Quincy

N could occupy (2) or (3); so, should be on the immediate right of K or M.

16) Qunicy is placed adjacent to Laxmi and Rehana. If there is only one person between Nini and Pallavi, then who should be that person?

- a) Kavita
- b) Munni
- c) Oliver
- d) Quincy

Ans: a

Explanation:

- 1- Laxmi
- 2- Munni
- 3- Nini
- 4- Kavita
- 5- Pallavi
- 7- Rehana
- 8- Qunicy

Thus, Kavita is between Nini and Pallavi.

Cubes 17-20:

The following questions are based on the information given below:

A cuboid shaped wooden block has 6 cm length, 4 cm breadth, and 1 cm height.

Two faces measuring 4 cm x 1 cm are colored in black.

Two faces measuring 6 cm x 1 cm are colored in red.

Two faces measuring 6 cm x 4 cm are colored in green.

The block is divided into 6 equal cubes of side 1 cm (from 6 cm side), 4 equal cubes of side 1 cm (from 4 cm side).

17) How many cubes having red, green and black colours on at least one side of the cube will be formed?

- a) 16
- b) 12
- c) 10
- d) 4

Ans: d

Explanation:

Such cubes are related to the corners of the cuboid.

Since the number of corners of the cuboid is 4.

18) How many small cubes will be formed?

- a) 6
- b) 12
- c) 16
- d) 24

Ans: d

Explanation:

Number of small cubes =  $l \times b \times h = 6 \times 4 \times 1 = 24$

19) How many cubes will have 4 colored sides and two non-colored sides?

- a) 4
- b) 8

c) 16

d) 10

Ans: a

Explanation:

Only 4 cubes situated at the corners of the cuboid will have 4 colored and 2 non-colored sides.

20) How many cubes will have green colour on two sides and rest of the four sides having no colour?

a) 12

b) 10

c) 4

d) 8

Ans: d

Explanation:

There are 16 small cubes attached to the outer walls of the cuboid. Therefore remaining inner small cubes will be the cubes having two sides green colored. So the required number =  $24 - 16 = 8$

Cubes 21-23:

Each of the six faces of a cube is painted in exactly one of the four colors, A, B, C and D, where A, B, C, D corresponds to different colors such that each face is at least one face that is painted in each color. The cube is then placed on a table and cut into 60 identical small cuboids by making the least possible number of cuts, each cut being parallel to some face of the cube.

21) What is the least possible number of small cuboids which have no face painted A?

a) 15

b) 20

c) 24

d) 40

Ans: a

Explanation:

The given cube is cut into 60 identical cuboids with least number of cuts. This can be done when the dimensions of the cube are  $3 \times 4 \times 5$ . Hence the number of cuts is  $2 + 3 + 4 = 9$ .

Now, we also need to select which 3 sides to paint with A, as the number of cuboids on each side are different. Naturally, we need to choose the three sides with the largest number of cuboids. This can be found on the two  $5 \times 4$  faces and the  $5 \times 3$  face. Also, these three faces are adjacent to each other. The two  $5 \times 4$  faces will each contain 20 cuboids, a total of 40 cuboids. These faces lie exactly opposite each other.

The  $5 \times 3$  face will be between these two faces. This face has 15 cuboids out of which 10 are already accounted for. Hence this side gives 5 more cuboids painted in white. The total number of cuboids painted in white is hence 45. Therefore, a total of 15 ( $60 - 45$ ) cuboids are not painted in white and this is the lowest such number.

22) What is the maximum possible number of small cuboids which have more than one face painted in the same color?

a) 9

b) 10

c) 12

d) 14

Ans: b

Explanation:

In the figure, the green colored highlighted area contains cuboids which have the same color lets say A. The other faces before it was cut are B, C, and D each with other different colors.

23) What is the maximum possible number of small cuboids which have all the sides painted?

a) 10

b) 8

c) 0

d) 6

Ans: c

Explanation:

No cuboid can have all the faces painted.

Cubes 24-27:

There are 128 cubes with me which are colored according to two schemes viz.

64 cubes each having two red adjacent faces and one yellow and other blue on their opposite faces while green on the rest.

64 cubes each having two adjacent blue faces and one red and other green on their opposite faces, while red on the rest. They are then mixed up.

24) How many cubes have at least two colored red faces each?

a) 0

b) 32

c) 64

d) 128

Ans: d

Explanation:

64 and 64 cubes of both types of cubes are such who have at least two colored faces red each. Therefore, the total number of the required cubes is 128.

25) What is the total number of red faces?

a) 0

b) 64



c) 320

d) 128

Ans: c

Explanation:

No. of red faces among first 64 cubes  $= 64 \times 2 = 128$

No. of red faces among second 64 cubes  $= 64 \times 3 = 192$

Therefore, total number of red faces  $= 128 + 192 = 320$

26) How many cubes have only one red face each?

a) 128

b) 32

c) 0

d) None

Ans: a

Explanation:

Out of 128 cubes, no cube has only one face as red.

27) Which two colour have the same number of faces?

a) Red and Yellow

b) Blue and Green

c) Red and Green

d) Red and Blue

Ans: b

Explanation:

First 64 cubes are such each of whose two faces are green and second 64 cubes are such each of whose two faces are blue. Therefore, green and blue colors have the same number of faces.

28) In a survey of university students, 64 had taken the mathematics course, 94 had taken the chemistry course, 58 had taken the physics course, 28 had taken mathematics and physics, 26 had taken mathematics and chemistry, 22 had taken chemistry and physics course, and 14 had taken all the three courses. Find how many had taken one course only.

- a) 106
- b) 120
- c) 123
- d) 96

Ans: a

Explanation:

Number of students who had taken only Mathematics =  $64 - [28 + 26 - 14] = 24$

Number of students who had taken only Chemistry =  $94 - [26 + 22 - 14] = 60$

Number of students who had taken only Physics =  $58 - [28 + 22 - 14] = 22$

Total students who had taken only one course =  $24 + 22 + 60 = 106$

Venn Diagrams 29-32:

In the following figure, small square represents the persons who know English, triangle to those who know Marathi, big square to those who know Telugu and circle to those who know Hindi. In the different regions of the figures from 1 to 12 are given.

29) How many persons can speak English and Hindi both the languages only?

- a) 5
- b) 8

c) 7

d) 18

Ans: a

Explanation:

The number of persons who can speak English and Hindi both only is 5.

30) How many persons can speak Marathi and Telugu both?

a) 10

b) 11

c) 13

d) None

Ans: c

Explanation:  $6+7 = 13$

31) How many persons can speak only English?

a) 9

b) 12

c) 7

d) 19

Ans: b

Explanation:

The number of persons who can speak English is 12.

32) How many persons can speak all the languages?

- a) 1
- b) 8
- c) 2
- d) None

Ans: d

Explanation:

There is no such person who can speak all the languages.

33) The marked price of a radio is Rs 1,600. Rahul gives a successive discount of 10%, r% to the Pradeep. If Pradeep pays Rs 1.224 for the radio, find the value of r.

- a) 10%
- b) 20%
- c) 25%
- d) 15%

Ans: d

Explanation:

Marked price of the article = Rs 1,600

Therefore, Selling price =  $(100 - 10)\% \times (100 - r\%) \times 1600$

$$= (90/100) \times [(100 - r) / 100] \times 1600$$

$$\text{Given, } 1224 = 9/10 \times (100 - r) \times 16$$

$$\Rightarrow 1224 \times 10 / (9 \times 16) = (100 - r)$$

$$85 = 100 - r$$

$$r = 15\%$$

34) On selling a pencil at 5% loss and a book at 15% gain, Kiran gains Rs 7. If he sells the pencil at 5% gain and the book at 10% gain, then he gains Rs 13. The actual price of the book is:

- a) 100
- b) 80
- c) 90
- d) 400

Ans: c

Explanation:

Let actual price of the book = Rs X

Let actual price of the pen = Rs Y

Therefore,  $(X + 15\% \text{ of } X) + (Y - 5\% \text{ of } Y) = X + Y + 7$

$$15X - 5Y = 700 \text{ .(i)}$$

Also  $(X + 10\% \text{ of } X) + (Y + 5\% \text{ of } Y) = X + Y + 13$

$$10X + 5Y = 1300$$

Using (i) and (ii) we get  $X = 80$ ,  $Y = 100$

Therefore, actual price of the book = Rs 80

35) A shopkeeper marks the price of his goods at 25% higher than the original price. After that, he allows a discount of 12%. What profit or loss does he get?

- a) 10% Profit
- b) 15% profit
- c) 10% Loss
- d) 15% Loss

Ans: a

Explanation:

Here,  $x = 25$  and  $y = -12$

Therefore, the net % change in value

$$= (x + y + xy/100) \%$$

$$= [25 - 12 + (-12 \times 25)/100]\% \text{ or } 10\%$$

Since the sign is positive, there is a profit of 10%

---

## Verbal Ability

### Source: TATA ELXSI\Verbal.txt

Tata Elxsi verbal questions with solutions are give here. These are the most repeated Tata Elxsi verbal questions which were asked in previous drive.

Tata Elxsi verbal section consists of 25 questions with a time duration of 25 minutes. In this section, questions related to articles and reading comprehension will be asked. For more details about Tata Elxsi pattern and syllabus - check here.

The recruitment process of Tata Elxsi keeps varying depending on the role. Firstly get a clear idea about the recruitment process for the role applied and then start your preparation with us.

Tata Elxsi Recruitment process

Now start your preparation for Tata Elxsi verbal questions.

1) They usually spend their holidays in \_\_\_\_\_ mountains.

- a) a
- b) an
- c) the
- d) no article

Ans: c

2) Ive been waiting for \_\_\_\_\_ long time.

- a) a
- b) an

- c) the
- d) no article

Ans: a

3) He hopes to join \_\_\_\_\_ university soon.

- a) a
- b) an
- c) the
- d) no article

Ans: a

4) Someone call \_\_\_\_\_ policeman!

- a) a
- b) an
- c) the
- d) no article

Ans: a

5) I haven't seen him in \_\_\_\_\_ five years.

- a) a
- b) an
- c) the
- d) no article

Ans: d



6) This is \_\_\_\_\_ best Mexican restaurant in the country.

- a) a
- b) an
- c) the
- d) no article

Ans: c

7) India will become \_\_\_\_\_ superpower shortly.

- a) a
- b) an
- c) the
- d) no article

Ans: a

8) Ram was \_\_\_\_\_ best student in the class.

- a) a
- b) an
- c) the
- d) no article

Ans: c

9) Please give me \_\_\_\_\_ useful gift.

- a) a
- b) an

- c) the
- d) no article

Ans: a

10) \_\_\_\_\_ Pandiyan Express is very popular.

- a) a
- b) an
- c) the
- d) no article

Ans: c

#### Reading Comprehension:

Directions for questions 11 to 15 - Read the short passage below and answer the questions that follow:

A sanctuary may be defined as a place where Man is passive and the rest of Nature active. Till quite recently Nature had her own sanctuaries, where man either did not go at all or only as a tool-using animal in comparatively small numbers. But now, in this machinery age, there is no place left where man cannot go with overwhelming forces at his command. He can strangle to death all the nobler wildlife in the world today. To-morrow he certainly will have done so, unless he exercises due foresight and self-control in the meantime.

There is not the slightest doubt that birds and mammals are now being killed off much faster than they can breed. And it is always the largest and noblest forms of life that suffer the most. The whales and elephants, lions and eagles, go. The rats and flies and all mean parasites remain. This is inevitable in certain cases. But it is wanton killing off that I am speaking of to-night.

Civilized man begins by destroying the very forms of wildlife he learns to appreciate most when he becomes still more civilized. The obvious remedy is to begin conservation at an earlier stage, when it is easier and better in every way, by enforcing laws for close seasons, game preserves, the selective protection of certain species, and sanctuaries.

I have just defined a sanctuary as a place where man is passive and the rest of Nature active. But this general definition is too absolute for any special case. The mere fact that man has to protect a sanctuary does away with his purely passive attitude. Then, he can be beneficially active by destroying pests and parasites, like bot-flies or mosquitoes, and

by finding antidotes for diseases like the epidemic which periodically kills off the rabbits and thus starves many of the carnivora to death. But, except in cases where the experiment has proved his intervention to be beneficial, the less he upsets the balance of Nature the better, even when he tries to be an earthly Providence.

11) The author implies that his first definition of a sanctuary is:

- A) Totally wrong
- B) Somewhat idealistic
- C) unhelpful
- D) indefensible
- E) immutable

Your Answer Options:

- 1) A
- 2) B
- 3) C
- 4) D
- 5) E

Ans: Option 2

Explanation:

The author says in sentence two that his previous definition was too absolute. Yet he admits that the less man upsets the balance of Nature the better. Hence his definition is not entirely right (because it is too idealistic) but it is not entirely wrong either. It is now easy to eliminate answers A, C, D, and E because they are strongly negative.

12) The authors argument that destroying bot-flies and mosquitoes would be a beneficial action is most weakened by all of the following except

- A) parasites have an important role to play in the regulation of populations
- B) the elimination of any species can have unpredictable effects on the balance of nature

- C) the pests themselves are part of the food chain
- D) these insects have been introduced to the area by human activities
- E) elimination of these insects would require the use of insecticides that kill a wide range of insects

Your Answer Options:

- 1) B
- 2) C & D
- 3) A
- 4) D
- 5) E

Ans: Option 4

Explanation:

In except questions find the true statements first. In this case, find four statements that weaken the idea that destroying pests is beneficial. That means finding statements that show that it is not a good idea to destroy the pests. Answers A, B, C, and E give reasons why destroying these insects might be a mistake. D, however, is the right answer because it suggests that eliminating these insects might not be wrong since they are not even natural inhabitants of the area.

13) It can be inferred that the passage is:

- a) part of an article in a scientific journal
- b) extracted from the minutes of a nature club
- c) part of a speech delivered to an educated audience
- d) a speech delivered in a court of law
- e) from a polemical article published in a magazine

Your Answer Options:

- 1) D & E
- 2) A
- 3) C
- 4) B
- 5) D

Ans: Option 3

Explanation:

From the words I am speaking of tonight we can infer that the words were delivered orally, and not during the day time. The only possible answer is C.

14) What should be the most appropriate central idea of this passage:

- A) The author argues that man kills big animals but saves mosquitoes & other parasites.
- B) Man is selfish by nature so he is up against the wildlife which is harmful for his survival
- C) Ecological balance, if not maintained by man will be harmful in long run.
- D) Author proposes a programme for not disturbing the balance of nature as it is beneficial for mankind.
- E) In view of the author, man should not intervene in natural environments.

Your Answer Options

- 1) A
- 2) D
- 3) C
- 4) E
- 5) B

Ans: Option 3

Explanation:

The author in his last paragraph warns the man against his attitude and suggests not to create disturbance in the ecological system and should have foresight for his own benefit.

15) Tone of the Author as expressed in the passage can be best described

- A) Descriptive to analytical
- B) Sarcastically humorous
- C) Objective to narrative
- D) Sarcastically critical to suggestive
- E) Ironically sarcastic to negative

Your Answer Options

- 1) A
- 2) C
- 3) D
- 4) E
- 5) B

Ans: Option 3

Explanation:

Very first paragraph determines the tone which goes on to criticise the human activities against nature sarcastically yet issues a suggestive warning to exercise due foresight and self-control. Entire passage revolves around this tone and concludes with the suggestion not to disturb the ecological balance.

Directions for questions 16 to 20 - Read the short passage below and answer the questions that follow:

Marie was born in 1867 in Warsaw, Poland, where her father was a Professor of Physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women. Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master's degree and doctorate in physics.

Marie was fortunate to have studied at the Sorbonne with some of the greatest scientists of her day, one of whom was Pierre Curie. Marie and Pierre were married in 1895 and spent many productive years working together in the physics laboratory. A short time after they discovered radium, Pierre was killed by a horse-drawn wagon in 1906. Marie was stunned by this horrible misfortune and endured heartbreaking anguish. Despondently she recalled their close relationship and the joy that they had shared in scientific research. The fact that she had two young daughters to raise by herself greatly increased her distress.

Curie's feeling of desolation finally began to fade when she was asked to succeed her husband as a physics professor at the Sorbonne. She was the first woman to be given a professorship at the world-famous university. In 1911 she received the Nobel Prize in chemistry for isolating radium. Although Marie Curie eventually suffered a fatal illness from her long exposure to radium, she never became disillusioned about her work. Regardless of the consequences, she had dedicated herself to science and to revealing the mysteries of the physical world.

16) Marie had a bright mind and a \_\_\_\_\_ personality.

(A) Strong (B) lighthearted (C) Humorous (D) Strange

Ans: B

17) When she learned that she could not attend the university in Warsaw, she felt \_\_\_\_\_ .

(A) Hopeless (B) Annoyed (C) Depressed (D) Worried

Ans: B

18) Marie \_\_\_\_\_ by leaving Poland and traveling to France to enter the Sorbonne.

(A) Challenged authority (B) Showed intelligence

(C) Behaved (D) Was distressed

Ans: A

19) \_\_\_\_\_ she remembered their joy together.

(A) Dejectedly (B) Worried (C) Tearfully (D) Happily

Ans: A

20) Her \_\_\_\_\_ began to fade when she returned to the Sorbonne to succeed her husband.

(A) Misfortune (B) Anger (C) Wretchedness (D) Disappointment

Ans: C

21) Even though she became fatally ill from working with radium, Marie Curie was never

(A) Troubled (B) Worried (C) Disappointed (D) Sorrowful

Ans: C

22) \_\_\_\_ teachers of this school are punctual.

a) a

b) an

c) the

d) no article

Ans: c

Explanation

Talks of a specific noun, hence the.

23) My book has become \_\_\_\_\_ best seller in academic English preparation.

a) a

b) an

c) the



d) no article

Ans: c

Explanation:

Talks about a specific condition; there cannot be more than one noun when we use words like best, ideal, perfect, absolute, only, hence the.

24) Can anyone give me \_\_\_\_\_ hand?

a) a

b) an

c) the

d) no article

Ans: a

Explanation:

Doesn't mention any specific noun, hence a. Giving a hand means helping someone.

25) \_\_\_\_\_ scheme failed for want of involvement.

a) a

b) an

c) the

d) no article

Ans: c

Explanation:

Talks of a specific scheme, hence the.

Now that you know what questions would come in Tata Elxsi verbal section, similarly prepare for other sections too.

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## Technical Aptitude

### Source: Accenture\Que 1 - Rat Count House\Problem Statement.txt

Question 1: Rat Count House

(Asked in Accenture OnCampus 10 Aug 2021, Slot 1)

Problem Description :

The function accepts two positive integers  $r$  and  $unit$  and a positive integer array  $arr$  of size  $n$  as its argument  $r$  represents the number of rats present in an area,  $unit$  is the amount of food each rat consumes and each  $i$ th element of array  $arr$  represents the amount of food present in  $i+1$  house number, where  $0 \leq i$

Note:

Return -1 if the array is null

Return 0 if the total amount of food from all houses is not sufficient for all the rats.

Computed values lie within the integer range.

Example:

Input:

$r$ : 7  
 $unit$ : 2  
 $n$ : 8  
 $arr$ : 2 8 3 5 7 4 1 2

Output:

4

Explanation:

Total amount of food required for all rats =  $r * unit$

=  $7 * 2 = 14$ .

The amount of food in 1st houses =  $2+8+3+5 = 18$ . Since, amount of food in 1st 4 houses is sufficient for all the rats. Thus, output is 4.

### Source: Accenture\Que 10\Problem Statement.txt

Question:10

(Asked in Accenture Offcampus 1 Aug 2021, Slot 2)

Problem Statement

A carry is a digit that is transferred to left if sum of digits exceeds 9 while adding two numbers from right-to-left one digit at a time

You are required to implement the following function.

```
Int NumberOfCarries(int num1 , int num2);
```

The function accepts two numbers ?num1? and ?num2? as its arguments. You are required to calculate and return the total number of carries generated while adding digits of two numbers ?num1? and ? num2?.

Assumption: num1, num2>=0

Example:

Input

Num 1: 451

Num 2: 349

Output

2

Explanation:

Adding ?num 1? and ?num 2? right-to-left results in 2 carries since ( 1+9) is 10. 1 is carried and (5+4=1) is 10, again 1 is carried. Hence 2 is returned.

Sample Input

Num 1: 23

Num 2: 563

Sample Output

0

**Source: Accenture\Que 11\Problem Statement.txt**

Question:11

(Asked in Accenture Offcampus 1 Aug 2021, Slot 3)

Problem Statement

You are given a function,

```
Void *ReplaceCharacter(Char str[], int n, char ch1, char ch2);
```

The function accepts a string ? str? of length n and two characters ?ch1? and ?ch2? as its arguments . Implement the function to modify and return the string ? str? in such a way that all occurrences of ?ch1? in original string are replaced by ?ch2? and all occurrences of ?ch2? in original string are replaced by ?ch1?.

Assumption: String Contains only lower-case alphabetical letters.

Note:

Return null if string is null.

If both characters are not present in string or both of them are same , then return the string unchanged.

Example:

Input:

Str: apples

ch1:a

ch2:p

Output:

paales

Explanation:

?A? in original string is replaced with ?p? and ?p? in original string is replaced with ?a?, thus output is paales.

### Source: Accenture\Que 12\Problem Statement.txt

Question:13

(Asked in Accenture Offcampus 2 Aug 2021, Slot 2)

#### Problem Statement

You are given a function,

Int MaxExponents (int a , int b);

You have to find and return the number between ?a? and ?b? ( range inclusive on both ends) which has the maximum exponent of 2.

The algorithm to find the number with maximum exponent of 2 between the given range is

Loop between ?a? and ?b?. Let the looping variable be ?i?.

Find the exponent (power) of 2 for each ?i? and store the number with maximum exponent of 2 so far in a variable , let say ?max?. Set ?max? to ?i? only if ?i? has more exponent of 2 than ?max?.

Return ?max?.

Assumption:  $a < b$

Note: If two or more numbers in the range have the same exponents of 2 , return the small number.

#### Example

Input:

7

12

Output:

8

Explanation:

Exponents of 2 in:

7-0

8-3

9-0

10-1

11-0

12-2

Hence maximum exponent if two is of 8.

**Source: Accenture\Que 13\Problem Statement.txt**

Question:12

(Asked in Accenture Offcampus 2 Aug 2021, Slot 1)

**Problem Statement**

You are required to implement the following function.

Int OperationChoices(int c, int n, int a , int b )

The function accepts 3 positive integers ?a? , ?b? and ?c ? as its arguments. Implement the function to return.

( a+ b ) , if c=1

( a ? b ) , if c=2

( a \* b ) , if c=3

(a / b) , if c =4

Assumption : All operations will result in integer output.

Example:

Input

c :1

a:12

b:16

Output:

Since ?c?=1 , (12+16) is performed which is equal to 28 , hence 28 is returned.

Sample Input

c : 2

a : 16

b : 20

Sample Output

-4

---

**Source: Accenture\Que 14\Problem Statement.txt**

Question : 14

(Asked in Accenture Offcampus 2 Aug 2021, Slot 3)

Problem Statement

You are required to implement the following function:

Int Calculate(int m, int n);

The function accepts 2 positive integer  $m$  and  $n$  as its arguments. You are required to calculate the sum of numbers divisible both by 3 and 5, between  $m$  and  $n$  both inclusive and return the same.

Note

$0 < m \leq n$

Example

Input:

$m : 12$

$n : 50$

Output

90

Explanation:

The numbers divisible by both 3 and 5, between 12 and 50 both inclusive are {15, 30, 45} and their sum is 90.

Sample Input

$m : 100$

$n : 160$

Sample Output

510

---

**Source: Accenture\Que 15\Problem Statement.txt**

Question 15

Problem Statement

You are required to input the size of the matrix then the elements of matrix, then you have to divide the main matrix in two sub matrices (even and odd) in such a way that element at 0 index will be considered as even and element at 1st index will be considered as odd and so on. then you have sort the even and odd matrices in ascending order then print the sum of second largest number from both the matrices

Example

enter the size of array : 5  
 enter element at 0 index : 3  
 enter element at 1 index : 4  
 enter element at 2 index : 1  
 enter element at 3 index : 7  
 enter element at 4 index : 9  
 Sorted even array : 1 3 9  
 Sorted odd array : 4 7

7

### Source: Accenture\Que 16\Problem Statement.txt

Question : 16

Instructions: You are required to write the code. You can click on compile and run anytime to check compilation/execution status. The code should be logically/syntactically correct.

Problem: Write a program in C to display the table of a number and print the sum of all the multiples in it.

Test Cases:

Test Case 1:

Input:

5

Expected Result Value:

5, 10, 15, 20, 25, 30, 35, 40, 45, 50

275

Test Case 2:

Input:

12

Expected Result Value:

12, 24, 36, 48, 60, 72, 84, 96, 108, 120

660

### Source: Accenture\Que 17\Problem Statement.txt

Question : 17

Instructions: You are required to write the code. You can click on compile and run anytime to check compilation/execution status. The code should be logically/syntactically correct.



Question: Write a program in C such that it takes a lower limit and upper limit as inputs and print all the intermediate palindrome numbers.

Test Cases:

TestCase 1:

Input :

10 , 80

Expected Result:

11 , 22 , 33 , 44 , 55 , 66 , 77.

Test Case 2:

Input:

100,200

Expected Result:

101 , 111 , 121 , 131 , 141 , 151 , 161 , 171 , 181 , 191.

### Source: Accenture\Que 18\Problem Statement.txt

Question : 18

Instructions: You are required to write the code. You can click on compile & run anytime to check the compilation/ execution status of the program. The submitted code should be logically/syntactically correct and pass all the test cases.

Ques: The program is supposed to calculate the distance between three points.

For

$x_1 = 1$   $y_1 = 1$

$x_2 = 2$   $y_2 = 4$

$x_3 = 3$   $y_3 = 6$

Distance is calculated as :  $\sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

### Source: Accenture\Que 19\Problem Statement.txt

Question : 19

Find the maximum value and its index in the array

Problem Statement :

You are given a function, void MaxInArray(int arr[], int length); The function accepts an integer array ?arr? of size ?length? as its argument. Implement the function to find the maximum element of the array and print the maximum element and its index to the standard output

(STDOUT). The maximum element and its index should be printed in separate lines.

Note:

Array index starts with 0

Maximum element and its index should be separated by a line in the output

Assume there is only 1 maximum element in the array

Print exactly what is asked, do not print any additional greeting messages

Example:

Input:

23 45 82 27 66 12 78 13 71 86

Output:

86

9

Explanation:

86 is the maximum element of the array at index 9.

### Source: Accenture\Que 2\Problem Statement.txt

Question 2:

(Asked in Accenture OnCampus 10 Aug 2021, Slot 2)

Problem Description :

The Binary number system only uses two digits, 0 and 1 and number system can be called binary string. You are required to implement the following function:

```
int OperationsBinaryString(char* str);
```

The function accepts a string str as its argument. The string str consists of binary digits eparated with an alphabet as follows:

? A denotes AND operation

? B denotes OR operation

? C denotes XOR Operation

You are required to calculate the result of the string str, scanning the string to right taking one opearation at a time, and return the same.

Note:

No order of priorities of operations is required

Length of str is odd

If str is NULL or None (in case of Python), return -1

Input:

str: 1C0C1C1A0B1

Output:

1

Explanation:

The alphabets in str when expanded becomes  $1 \text{ XOR } 0 \text{ XOR } 1 \text{ XOR } 1 \text{ AND } 0 \text{ OR } 1$ , result of the expression becomes 1, hence 1 is returned.

Sample Input:

0C1A1B1C1C1B0A0

Output:

0

### Source: Accenture\Que 20 - Autobiographical Number\Problem Statement.txt

Question : 20

Autobiographical Number

Problem Statement :

An Autobiographical Number is a number N such that the first digit of N represents the count of how many zeroes are there in N, the second digit represents the count of how many ones are there in N and so on.

You are given a function, def FindAutoCount(n):

The function accepts string ?n? which is a number and checks whether the number is an autobiographical number or not. If it is, an integer is returned, i.e. the count of distinct numbers in ?n?. If not, it returns 0.

Assumption:

The input string will not be longer than 10 characters.

Input string will consist of numeric characters.

Note:

If string is None return 0.

Example:

Input:

n: ?1210?

Output:

3

Explanation:

0th position in the input contains the number of 0 present in input, i.e. 1, in 1st position the count of number of 1s in input i.e. 2, in 2nd position the count of 2s in input i.e. 1, and in 3rd position the count of 3s i.e. 0, so the number is an autobiographical number.

Now unique numbers in the input are 0, 1, 2, so the count of unique numbers is 3. So 3 is returned.

### Source: Accenture\Que 3 - Password Checker\Problem Statement.txt

Question 3: Password Checker

(Asked in Accenture OnCampus 10 Aug 2021, Slot 3)

You are given a function.

```
int CheckPassword(char str[], int n);
```

The function accepts string str of size n as an argument. Implement the function which returns 1 if given string str is valid password else 0.

str is a valid password if it satisfies the below conditions.

- ? At least 4 characters
- ? At least one numeric digit
- ? At Least one Capital Letter
- ? Must not have space or slash (/)
- ? Starting character must not be a number

Assumption:

Input string will not be empty.

Example:

Input:

aA1\_67

Output:

1

Sample Input:

a987 abC012

Output:

0

### Source: Accenture\Que 4\Problem Statement.txt

Question 4:

(Asked in Accenture OnCampus 11 Aug 2021, Slot 1)

You are given a function,  
`int findCount(int arr[], int length, int num, int diff);`

The function accepts an integer array `arr`, its length and two integer variables `num` and `diff`. Implement this function to find and return the number of elements of `arr` having an absolute difference of less than or equal to `diff` with `num`.

Note: In case there is no element in `arr` whose absolute difference with `num` is less than or equal to `diff`, return -1.

Example:

Input:

`arr: 12 3 14 56 77 13`

`num: 13`

`diff: 2`

Output:

3

Explanation:

Elements of `arr` having absolute difference of less than or equal to `diff` i.e. 2 with `num` i.e. 13 are 12, 13 and 14.

### Source: Accenture\Que 5\Problem Statement.txt

Question 5 :

(Asked in Accenture OnCampus 11 Aug 2021, Slot 2)

Implement the following Function

`def differenceofSum(n, m)`

The function accepts two integers `n`, `m` as arguments Find the sum of all numbers in range from 1 to `m`(both inclusive) that are not divisible by `n`. Return difference between sum of integers not divisible by `n` with sum of numbers divisible by `n`.

Assumption:

`n > 0` and `m > 0`

Sum lies between integral range

Example

Input

`n:4`

`m:20`

Output

90

Explanation

Sum of numbers divisible by 4 are  $4 + 8 + 12 + 16 + 20 = 60$

Sum of numbers not divisible by 4 are  $1 + 2 + 3 + 5 + 6 + 7 + 9 + 10 + 11 + 13 + 14 + 15 + 17 + 18 + 19 = 150$

Difference  $150 - 60 = 90$

Sample Input

n:3

m:10

Sample Output

19

### Source: Accenture\Que 6\Problem Statement.txt

Question:6

(Asked in Accenture OnCampus 11 Aug 2021, Slot 3)

You are required to implement the following Function

```
def LargeSmallSum(arr)
```

The function accepts an integers arr of size ?length? as its arguments you are required to return the sum of second largest element from the even positions and second smallest from the odd position of given ?arr?

Assumption:

All array elements are unique

Treat the 0th position as even

NOTE

Return 0 if array is empty

Return 0, if array length is 3 or less than 3

Example

Input

arr:3 2 1 7 5 4

Output

7

Explanation

Second largest among even position elements(1 3 5) is 3

Second smallest among odd position element is 4

Thus output is  $3+4 = 7$

Sample Input

arr:1 8 0 2 3 5 6

Sample Output

8

**Source: Accenture\Que 7\Problem Statement.txt**

Question:7

(Asked in Accenture OnCampus 12 Aug 2021, Slot 1)

Implement the following Function

```
def ProductSmallestPair(sum, arr)
```

The function accepts an integers sum and an integer array arr of size n. Implement the function to find the pair, (arr[j], arr[k]) where  $j \neq k$ , Such that arr[j] and arr[k] are the least two elements of array ( $\text{arr}[j] + \text{arr}[k] \leq \text{sum}$ ) and return the product of element of this pair

NOTE

Return -1 if array is empty or if  $n < 2$ 

Return 0, if no such pairs found

All computed values lie within integer range

Example

Input

sum:9

Arr:5 2 4 3 9 7 1

Output

2

Explanation

Pair of least two element is (2, 1)  $2 + 1 = 3 < 9$ , Product of (2, 1)  $2 * 1 = 2$ . Thus, output is 2

Sample Input

sum:4

Arr:9 8 3 -7 3 9

Sample Output

-21

**Source: Accenture\Que 8\Problem Statement.txt**

Question:8

(Asked in Accenture OnCampus 12 Aug 2021, Slot 2)

N-base notation is a system for writing numbers that uses only n different symbols, This symbols are the first n symbols

from the given notation list(Including the symbol for o) Decimal to n base notation are (0:0, 1:1, 2:2, 3:3, 4:4, 5:5, 6:6, 7:7, 8:8, 9:9, 10:A,11:B and so on upto 35:Z)

Implement the following function

Char\* DectoNBase(int n, int num):

The function accept positive integer n and num Implement the function to calculate the n-base equivalent of num and return the same as a string

Steps:

Divide the decimal number by n,Treat the division as the integer division

Write the the remainder (in n-base notation)

Divide the quotient again by n, Treat the division as integer division

Repeat step 2 and 3 until the quotient is 0

The n-base value is the sequence of the remainders from last to first

Assumption:

$1 < n \leq 36$

Example

Input

n: 12

num: 718

Output

4BA

Explanation

| num | Divisor | quotient | remainder |
|-----|---------|----------|-----------|
| 718 | 12      | 59       | 10(A)     |
| 59  | 12      | 4        | 11(B)     |
| 4   | 12      | 0        | 4(4)      |

Sample Input

n: 21

num: 5678

Sample Output



CI8

**Source: Accenture\Que 9\Problem Statement.txt**

Question:9

(Asked in Accenture Offcampus 1 Aug 2021, Slot 1)

Implement the following functions.a

```
char*MoveHyphen(char str[],int n);
```

The function accepts a string ?str? of length ?n?, that contains alphabets and hyphens (-). Implement the function to move all hyphens(-) in the string to the front of the given string.

NOTE:- Return null if str is null.

Example :-

Input:

str.Move-Hyphens-to-Front

Output:

?MoveHyphenstoFront

Explanation:-

The string ?Move-Hyphens -to-front? has 3 hyphens (-), which are moved to the front of the string, this output is ?? MoveHyphen?

Sample Input

Str: String-Compare

Sample Output-

-StringCompare

**Source: Cognizant\Que 1\Problem Statement.txt**

Problem Statement ? Write a program to calculate the fuel consumption of your truck.The program should ask the user to enter the quantity of diesel to fill up the tank and the distance covered till the tank goes dry.Calculate the fuel consumption and display it in the format (liters per 100 kilometers).

Convert the same result to the U.S. style of miles per gallon and display the result. If the quantity or distance is zero or negative display ? is an Invalid Input?.

[Note: The US approach of fuel consumption calculation (distance / fuel) is the inverse of the European approach (fuel / distance ). Also note that 1 kilometer is 0.6214 miles, and 1 liter is 0.2642 gallons.]

The result should be with two decimal place.To get two decimal place refer the below-mentioned print statement :

```
float cost=670.23;
```

```
System.out.printf("You need a sum of Rs.%.2f to cover the trip?",cost);
```

Sample Input 1:

```
Enter the no of liters to fill the tank
20
```

```
Enter the distance covered
150
```

Sample Output 1:

```
Liters/100KM
13.33
```

```
Miles/gallons
17.64
```

Explanation:

For 150 KM fuel consumption is 20 liters,  
 Then for 100 KM fuel consumption would be  $(20/150)*100=13.33$ ,  
 Distance is given in KM, we have to convert it to miles  $(150*0.6214)=93.21$ ,  
 Fuel consumption is given in liters, we have to convert it to gallons  $(20*0.2642)=5.284$ ,  
 Then find  $(\text{miles/gallons})=(93.21/5.284)=17.64$   
 Sample Input 2:

```
Enter the no of liters to fill the tank
-5
```

Sample Output 2:

-5 is an Invalid Input

Sample Input 3:

```
Enter the no of liters to fill the tank
25
```

```
Enter the distance covered
-21
```

Sample Output 3:

-21 is an Invalid Input

**Source: Cognizant\Que 10\Problem Statement.txt**

Problem Statement ? Chaman planned to choose a four digit lucky number for his car. His lucky numbers are 3,5 and 7.

Help him find the number, whose sum is divisible by 3 or 5 or 7. Provide a valid car number, Fails to provide a valid input then display that number is not a valid car number.

Note : The input other than 4 digit positive number[includes negative and 0] is considered as invalid.

Refer the samples, to read and display the data.

Sample Input 1:

Enter the car no:1234

Sample Output 1:

Lucky Number

Sample Input 2:

Enter the car no:1214

Sample Output 2:

Sorry its not my lucky number

Sample Input 3:

Enter the car no:14

Sample Output 3:

14 is not a valid car number

### Source: Cognizant\Que 11\Problem Statement.txt

Problem Statement ?

IIHM institution is offering a variety of courses to students. Students have a facility to check whether a particular course is available in the institution. Write a program to help the institution accomplish this task. If the number is less than or equal to zero display ?Invalid Range?.

Assume maximum number of courses is 20.

Sample Input 1:

Enter no of course:

5

Enter course names:

Java

Oracle

C++

Mysql

Dotnet

Enter the course to be searched:

C++

Sample Output 1:

C++ course is available

Sample Input 2:

Enter no of course:

3

Enter course names:

Java

Oracle

Dotnet

Enter the course to be searched:

C++

Sample Output 2:

C++ course is not available

Sample Input 3:

Enter no of course:

0

Sample Output 3:

Invalid Range

### Source: Cognizant\Que 12\Problem Statement.txt

Problem Statement ? Mayuri buys ?N? no of products from a shop. The shop offers a different percentage of discount on each item. She wants to know the item that has the minimum discount offer, so that she can avoid buying that and save money.

[Input Format: The first input refers to the no of items; the second input is the item name, price and discount percentage separated by comma(,)]

Assume the minimum discount offer is in the form of Integer.

Note: There can be more than one product with a minimum discount.

Sample Input 1:

4

mobile,10000,20

shoe,5000,10

watch,6000,15

laptop,35000,5

Sample Output 1:

shoe

Explanation: The discount on the mobile is 2000, the discount on the shoe is 500, the discount on the watch is 900 and the discount on the laptop is 1750. So the discount on the shoe is the minimum.

Sample Input 2:

4

Mobile,5000,10

shoe,5000,10

WATCH,5000,10

Laptop,5000,10

Sample Output 2:

Mobile

shoe

WATCH

Laptop

**Source: Cognizant\Que 13\Problem Statement.txt**

Problem Statement ? Raj wants to know the maximum marks scored by him in each semester. The mark should be between 0 to 100 ,if goes beyond the range display ?You have entered invalid mark.?

Sample Input 1:

Enter no of semester:

3

Enter no of subjects in 1 semester:

3

Enter no of subjects in 2 semester:

4

Enter no of subjects in 3 semester:

2

Marks obtained in semester 1:

50

60

70

Marks obtained in semester 2:

90

98

76

67

Marks obtained in semester 3:

89

76

Sample Output 1:

Maximum mark in 1 semester:70

Maximum mark in 2 semester:98

Maximum mark in 3 semester:89

Sample Input 2:

Enter no of semester:

3

Enter no of subjects in 1 semester:

3

Enter no of subjects in 2 semester:

4

Enter no of subjects in 3 semester:

2

Marks obtained in semester 1:

55

67

98

Marks obtained in semester 2:

67

-98

Sample Output 2:

You have entered invalid mark.

---

**Source: Cognizant\Que 14\Problem Statement.txt**

Problem Statement ? Bela teaches her daughter to find the factors of a given number. When she provides a number to her daughter, she should tell the factors of that number. Help her to do this, by writing a program. Write a class FindFactor.java and write the main method in it.

Note :

If the input provided is negative, ignore the sign and provide the output. If the input is zero  
If the input is zero the output should be ?No Factors?.

Sample Input 1 :

54

Sample Output 1 :

1, 2, 3, 6, 9, 18, 27, 54

Sample Input 2 :

-1869

Sample Output 2 :

1, 3, 7, 21, 89, 267, 623, 1869

---

**Source: Cognizant\Que 15\Problem Statement.txt**

Here the user should give you two numbers for that you have to find the LCM of them,

Where LCM is Lowest Common factor of two numbers is the smallest number that divides both of them.

Sample Input 1:

10 15

Sample Output 1:

30

Sample Input 2:

60 20

Sample Output 2:

60

---

**Source: Cognizant\Que 16\Problem Statement.txt**

Ananya Pandey's professor has suggested her to prepare well for the lesson on seasons. While her professor says a month, she needs to tell the season corresponding to that month. Write a program to solve the above task.

Spring ? March to May,

Summer ? June to August,

Autumn ? September to November and,

Winter ? December to February.

The month should be in the range of 1 to 12. If not the output should be ?Invalid month?.

Sample Input 1:

Enter the month: 08

Sample Output 1:

Season: Summer

Sample Input 2:

Enter the month: 14

Sample Output 2:

Invalid month

---

**Source: Cognizant\Que 17\Problem Statement.txt**

There is a scheme declared in theatre, in which one ticket gets a 10% discount on the total cost of tickets if there is a bulk booking which is more than 20 tickets, and if there is a special coupon card then there is 2% discount on total the cost. Also, customer can buy Refreshments which additional cost of Rs. 50 per person.

The cost of the K class Rs. 75 and the Q class is Rs. 150.

Write a program to find the total cost as per the scheme.

Note: In-class K and class Q, you have to buy at a time the minimum of 5 tickets and a maximum of 40. If this condition fails then display the message ?Minimum of 5 and Maximum of 40 Tickets?.

If the user is given a value other than class ?K? or ?Q? the output should be ?Invalid Input?.

The cost should be printed correctly to the two decimal places.

Sample Input 1:

Enter the no of ticket:35

Do you want refreshment:y



Do you have coupon code:y

Enter the circle:k

Sample Output 1:

Ticket cost:4065.25

Sample Input 2:

Enter the no of ticket:1

Sample Output 2:

Minimum of 5 and Maximum of 40 Tickets

---

### Source: Cognizant\Que 18\Problem Statement.txt

Print the prime numbers from 1 to N, where N is an input.

Sample Input 1:

15

Sample Output 1:

2    3    5    7    11    13

Sample Input 2:

5

Sample Output 2:

2    3    5

---

### Source: Cognizant\Que 19\Problem Statement.txt

Bela teaches his son to find the factors of the given numbers. When he provides a number to his son, he should tell the factors of that number. Help him to do this, by writing a program.

Note:

If the input is zero (0) then the output will be "No Factors".

If the input is any negative number, then first convert it positive solve.

Sample Input 1:

54

Sample Output 1:

Factors for 56 are:

1

2

4

7

8

14

28

56

Sample Input 2:

-256

Sample Output 2:

Factors of 256 are:

1

2

4

8

16

32

64

128

256

---

**Source: Cognizant\Que 2\Problem Statement.txt**

Problem Statement ? Vohra went to a movie with his friends in a Wave theatre and during break time he bought pizzas, puffs and cool drinks. Consider the following prices :

Rs.100/pizza

Rs.20/puffs

Rs.10/cooldrink

Generate a bill for What Vohra has bought.

Sample Input 1:

Enter the no of pizzas bought:10

Enter the no of puffs bought:12

Enter the no of cool drinks bought:5

Sample Output 1:

Bill Details

No of pizzas:10

No of puffs:12

No of cooldrinks:5

Total price=1290

ENJOY THE SHOW!!!

### Source: Cognizant\Que 20\Problem Statement.txt

Problem statement:- Elliot made a KeyLogger for his friend Romero, so that he can see the passwords of his friend. Keylogger is a software that can tell you the buttons pressed in the keyboard without the consent of the user, and hence unethical. Elliot made it to hack Romero's passwords. The one problem is, Romero writes the passwords in lowercase characters only, and the keylogger only takes the values of the keys. Like, for a it takes 1, for b 2, and for z 26. For a given number Elliot produces all combinations of passwords in a dictionary and starts a dictionary based password attack. For a given number, print all the possible passwords in a lexicographic order.

Input Format:

One line, denoting the value given by the keylogger

Output Format:

All possible combinations of keyloggers in new lines are lexicographically ordered.

Constraints:

$2 \leq \text{Number of digit in input} \leq 1000$

Sample Input:

1234

Sample Output:

abcd

awd

lcd

Explanation:

For 12, you can take 1,2 that is ab, or you can take l.

---

**Source: Cognizant\Que 21\Problem Statement.txt**

Problem Statement- Ramesh went to a bookshop to buy books. There is a list of books with their value and price. Now Ramesh has limited money but he wants maximum value possible. Now there are 2 kinds of books, one is denoted with 1, that is independent, another one is denoted as 2, which you have to buy in double, that means you can not buy a single or odd number of those books.

Print the maximum value Ramesh can extract from the books.

Input Format:

First line contains two integers, n (Number of books) and T, total money he has.

Then n lines, 4 variables in each line,

The serial number of the book

The value of the book

The price of the book

The marking (1 or 2)

Output Format:

Maximum value possible to be bought.

Constraints:

Number of books:  $\leq 100$

Maximum Money with Ramesh  $\leq 1000$

Max price of a book  $\leq 1000$

Sample Input:

5 20

1 3 7 0

3 9 10 1

2 4 3 1

7 3 2 0

22 7 7 0

Sample Output:

20

Explanation:

It will be the 1st book, 2nd and the third book

### Source: Cognizant\Que 22\Problem Statement.txt

**Problem Statement-** A taxi can take multiple passengers to the railway station at the same time. On the way back to the starting point, the taxi driver may pick up additional passengers for his next trip to the airport. A map of passenger location has been created, represented as a square matrix.

The Matrix is filled with cells, and each cell will have an initial value as follows:

A value greater than or equal to zero represents a path.

A value equal to 1 represents a passenger.

A value equal to -1 represents an obstruction.

The rules of motion of taxi are as follows:

The Taxi driver starts at (0,0) and the railway station is at (n-1,n-1). Movement towards the railway station is right or down, through valid path cells.

After reaching (n-1,n-1) the taxi driver travels back to (0,0) by travelling left or up through valid path cells.

When passing through a path cell containing a passenger, the passenger is picked up. Once the rider is picked up, the cell becomes an empty path cell.

If there is no valid path between (0,0) and (n-1,n-1), then no passenger can be picked.

The goal is to collect as many passengers as possible so that the driver can maximize his earnings.

Sample Input 0

```
4
4
0 0 0 1
1 0 0 0
0 0 0 0
0 0 0 0
```

Sample Output 0

```
2
```

Explanation 0

The driver can contain a maximum of 2 passengers by taking the following path (0,0) ? (0,1) ? (0,2) ? (0,3) ? (1,3) ? (2,3) ? (3,3) ? (3,2) ? (3,1) ? (3,0) ? (2,0) ? (1,0) ? (0,0)

Sample Input 1

```
3
3
0 1 -1
1 0 -1
1 1 1
```

Sample Output 1

5

Explanation 1

The driver can contain a maximum of 5 passengers by taking the following path (0,0) ? (0,1) ? (1,1) ? (2,1) ? (2,2) ? (2,1) ? (2,0) ? (1,0) ? (0,0)

### Source: Cognizant\Que 23\Problem Statement.txt

Problem Statement ? Ratan is a crazy rich person. And he is blessed with luck, so he always made the best profit possible with the shares he bought. That means he bought a share at a low price and sold it at a high price to maximize his profit. Now you are an income tax officer and you need to calculate the profit he made with the given values of stock prices each day. You have to calculate only the maximum profit Ratan earned.

Note that:

Ratan never goes into loss.

Example 1

Price=[1,6,2]  
Ratan buys it on the first day and sells it on the second.

Example 2

Price=[9,8,6]  
The Price always went down, Ratan never bought it.

Input Format:

First line with an integer  $n$ , denoting the number days with the value of the stock  
Next  $n$  days, telling the price of the stock on that very day.

Output Format:

Maximum profit done by Ratan in a single line.

Constraints:

Number of days  $\leq 10^8$

Sample Input for Custom Testing

STDIN

????

7

1

9

2

11

1

9

2

Sample Output

10

Explanation

The maximum profit possible is when Ratan buys it in 1 rupees and sells it in 11.

---

**Source: Cognizant\Que 24\Problem Statement.txt**

Problem Statement ? Codu is given a string and he thinks the letters that are repeated do have more power. He gathers only the repeating characters and keeps them as the most powerful to least powerful manner. Now it is your turn to write a code that will help Codu to do that.

Note that: only lowercase alphabets are accepted in input.

Input Format:

A string in a single line

Output Format:

A string made of only the repeated characters as sorted their frequency reducing, if the same the lower ascii value comes before.

Constraints:

Length of string  $\leq 10^5$

Sample Input:

abcdefghaabc

Sample Output:

abc

---

**Source: Cognizant\Que 25\Problem Statement.txt**

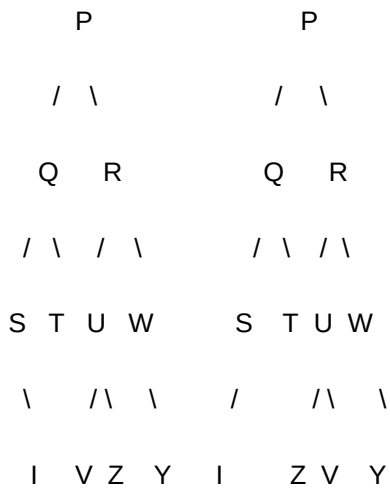
Problem Statement ? How will we represent a binary tree? We can use a bracket structure for all of the edges, like (Parentnode, Childnode). Now if we use a node in a child node more than once, the tree can not be valid. Same for the parent node, a node can not be taken more than twice in a graph as a parent node.

Suppose we see this one graph

(P,Q)(P,R)(Q,T)(R,W)(U,V)(Q,S)(R,U)(U,Z)(S,I)(W,Y)

A tree with those edges may be illustrated in many ways. Here are two:





The following is a recursive definition for the S-expression of a tree.

S-exp(node)=(node->val(S-exp(node->first\_child))(S-exp(node->second\_child))),if node

!NULL=??,node= =NULL

Where first\_child->val<second\_child->val(first\_child->val is lexicographically than second\_child->val)

This tree can be represented in S-expression in multiple ways.The lexicographically smallest way of expressing it as follows:

P(Q(S(I))(T))(R(U(V)(Z))(W(Y))))

Translate the node-pair representation into its lexicographically smallest S-expression or report any errors that do not conform to the definition of the binary tree.

The List of errors with their codes is as follows:

| Error         | Reason                                                         |
|---------------|----------------------------------------------------------------|
| Code Stopped1 | More than 2 children                                           |
| Code Stopped2 | Duplicate Edges                                                |
| Code Stopped3 | Cycle Present(node is direct descendant of more than one node) |
| Code Stopped4 | Multiple Roots                                                 |
| Code Stopped5 | Any other error                                                |

## Functional Description

Complete the function sExpression in the editor below.

The function must return either the lexicographically lowest S-expression or the lexicographically lowest error code as a string.

sExpression has the following parameter(s):

Nodes: a string of space-separated parenthetical elements, each of which contains the name of two nodes connected by a comma.

Constraints:

All node names are single characters in the range `ascii[A-Z]`.

The maximum node count is 26.

There is no specific order to the input (parent,child) pairs.

>Input Format for Custom Testing

>Sample Case 0

Sample Input 0

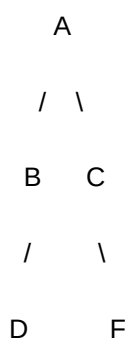
(B,D) (D,E) (A,B) (C,F) (E,G) (A,C)

Sample output 0

(A(B(D(E(G)))))(C(F))

Explanation 0

A representation of tree is as follows:



/

E

/

G

&gt;Sample Case 1

Input:

(C,E)(D,F)(A,B)(A,C)(B,K)

Output:

A(B(K))(C(E))D(F))

**Source: Cognizant\Que 26\Problem Statement.txt**

Problem Statement ? Sahil watches TV all day and gets bored. He started playing this dumb game of identifying minimum number of inputs needed to reach a channel. As his cousin, you have to help him, but you live far from his house. So you decide to write a code that will ask Sahil for some inputs and give outputs respectively.

Here are the problems you need to keep in mind,

There are 13 buttons on his remote: 10 buttons for the numbers (0-9) to form integers denoting respective channel index, ?Up channel? button and ?Down channel? button for going i +1th channel and i-1th channel from i respectively, and a ?Last viewed? button to see what?s the last channel before it.

The number buttons allow you to jump directly to a specific channel (Ex: to go to channel 172 by typing 1,7,2).

If the channel which you are in is ith and that is the max channel index possible, by Up channel, you will reach the first channel possible. Same goes for the down channel button. You can go to the highest channel possible if you go down from the lowest channel possible.

Sahil can get from one channel to the next in one of the two ways.

Sahil?s parents have set some parental control on some channels on Aniruth?s television. The ?Up Channel ? and ?Down buttons? buttons skip these channels as these channels are not viewable.

Given a list of channels to view, the lowest channel, the highest channel, and a list of blocked channels, your program should return the minimum number of clicks necessary to get through all the shows that Anirudh would like to match.

Input Format

First line is the lowest Channel

Second-line is the highest Channel

Followed by a number of blocked channels B,  
 and the next B lines contain the actual blocked channels.  
 Followed by the number of Channels to view V, and the next V lines contain the actual channels to view.

### Constraints

The lowest channel on the television will be greater than 0. and less than or equal to 10,000.  
 The highest channel on the television will be greater than or equal to the lowest channel. and less than or equal to 10,000.  
 The list of channels that are blocked on Anirudh's television. All the channels in this list will be valid channels (greater than or equal to lowest channel, less than or equal 1 to highest channel). Duplicates may be Ignored. The blocked list can be a maximum of 40 channels.  
 The sequence that Sahil must view contains between 1 and 50 elements. inclusive. All channels in this sequence are not in the blocked list and are between lowest channel and highest channel. Inclusive.

### Sample Input 0:

1  
 20  
 2  
 18  
 19  
 5  
 15  
 14  
 17  
 1  
 17

### Sample output 0:

7

---

### Source: Cognizant\Que 27\Problem Statement.txt

Problem Statement ? A company has a list of jobs to perform. Each job has a start time, end time and profit value. The manager has asked his employee Anirudh to pick jobs of his choice. Anirudh being greedy wants to select jobs for him in such a way that would maximize his earnings.

Given a list of jobs how many jobs and total earning are left for other employees once Anirudh

Picks jobs of his choice.

Note: Anirudh can perform only one job at a time.

Input format:

Each Job has 3 pieces of info ? Start Time,End Time and Profit

The first line contains the number of Jobs for the day. Say ?n?. So there will be ?3n lines following as each job has 3 lines.

Each of the next ?3n? lines contains jobs in the following format:

start\_time

end-time

Profit

start-time and end-time are in HHMM 24HRS format i.e. 9am is 0900 and 9PM is 2100

Constraints

The number of jobs in the day i.e? is less than 10000

$0 \leq n \leq 10000$

start-time is always less than end time.

Output format :-

Program should return an array of 2 integers where

1st one is number of jobs left and earnings of other employees

Sample Input 1 :

4

0200

0300

10

0400

0700

20

0300

0800

30

0900

1000

50

Sample Output 1:

1

20

Sample Explanation 1

Chooses 1st, 3rd and 4th job cause they don't overlap. So only second job is remaining.

Sample Input 2:

5

0805

0830

100

0835

0900

100

0905

0930

100

0935

1000

100

1005

1030

100

Sample output 2:

0

0

Sample Explanation 2:

Anirudh can work on all appointments as there are none overlapping.

Hence 0 appointments and 0 earnings for other employees.

**Source: Cognizant\Que 28\Problem Statement.txt**

Problem Statement : Given two integers, the task is to find the hamming distance between two integers. Hamming Distance between two integers is the number of bits that are different at the same position in both numbers.

Example :

Input 1:

a= 10 , b =8

Output 1:

1

Explanation: 10= 1010 , 8=1000 , No of different bits=1.

Input 2:

a=9, b=14

Output 2:

3

Explanation: 9=1001 , 14=1110 , No. of different bits=3.

---

**Source: Cognizant\Que 29\Problem Statement.txt**

Problem Statement: Given boolean 2D matrix , find the number of islands. A group of connected 1s form an island. For example, the below matrix contains 5 islands.

Example :

m =5, n=5

1 1 0 0 0

0 1 0 0 1

1 0 0 1 1

0 0 0 0 0

1 0 1 0 1

Output :

5

Explanation: There are total 5 island in the matrix.

---

**Source: Cognizant\Que 3\Problem Statement.txt**

Problem Statement ? Ritik wants a magic board, which displays a character for a corresponding number for his science project. Help him to develop such an application.

For example when the digits 65,66,67,68 are entered, the alphabet ABCD are to be displayed.

[Assume the number of inputs should be always 4 ]

Sample Input 1:

Enter the digits:

65

66

67

68

Sample Output 1:

65-A

66-B

67-C

68-D



Sample Input 2:

Enter the digits:

115

116

101

112

Sample Output 2:

115-s

116-t

101-e

112-p

### Source: Cognizant\Que 4\Problem Statement.txt

Problem Statement ? FOE college wants to recognize the department which has succeeded in getting the maximum number of placements for this academic year. The departments that have participated in the recruitment drive are CSE,ECE, MECH. Help the college find the department getting maximum placements. Check for all the possible output given in the sample snapshot

Note : If any input is negative, the output should be ?Input is Invalid?. If all department has equal number of placements, the output should be ?None of the department has got the highest placement?.

Sample Input 1:

Enter the no of students placed in CSE:90

Enter the no of students placed in ECE:45

Enter the no of students placed in MECH:70

Sample Output 1:

Highest placement

CSE

Sample Input 2:

Enter the no of students placed in CSE:55

Enter the no of students placed in ECE:85

Enter the no of students placed in MECH:85

Sample Output 2:

Highest placement

ECE

MECH

Sample Input 3:

Enter the no of students placed in CSE:0

Enter the no of students placed in ECE:0  
 Enter the no of students placed in MECH:0  
 Sample Output 3:

None of the department has got the highest placement  
 Sample Input 4:

Enter the no of students placed in CSE:10  
 Enter the no of students placed in ECE:-50  
 Enter the no of students placed in MECH:40  
 Sample Output 4:

Input is Invalid

### Source: Cognizant\Que 5\Problem Statement.txt

Problem Statement ? In a theater, there is a discount scheme announced where one gets a 10% discount on the total cost of tickets when there is a bulk booking of more than 20 tickets, and a discount of 2% on the total cost of tickets if a special coupon card is submitted. Develop a program to find the total cost as per the scheme. The cost of the k class ticket is Rs.75 and q class is Rs.150. Refreshments can also be opted by paying an additional of Rs. 50 per member.

Hint: k and q and You have to book minimum of 5 tickets and maximum of 40 at a time. If fails display ?Minimum of 5 and Maximum of 40 Tickets?. If circle is given a value other than ?k? or ?q? the output should be ?Invalid Input?.

The ticket cost should be printed exactly to two decimal places.

Sample Input 1:

Enter the no of ticket:35  
 Do you want refreshment:y  
 Do you have coupon code:y  
 Enter the circle:k  
 Sample Output 1:

Ticket cost:4065.25  
 Sample Input 2:

Enter the no of ticket:1  
 Sample Output 2:

Minimum of 5 and Maximum of 40 Tickets

### Source: Cognizant\Que 6\Problem Statement.txt

Problem Statement ? Rhea Pandey?s teacher has asked her to prepare well for the lesson on seasons. When her teacher tells a month, she needs to say the season corresponding to that month. Write a program to solve the above task.

Spring ? March to May,  
 Summer ? June to August,

Autumn ? September to November and,

Winter ? December to February.

Month should be in the range 1 to 12. If not the output should be ?Invalid month?.

Sample Input 1:

Enter the month:11

Sample Output 1:

Season:Autumn

Sample Input 2:

Enter the month:13

Sample Output 2:

Invalid month

### Source: Cognizant\Que 7\Problem Statement.txt

Problem Statement ? To speed up his composition of generating unpredictable rhythms, Blue Bandit wants the list of prime numbers available in a range of numbers.Can you help him out?

Write a java program to print all prime numbers in the interval [a,b] (a and b, both inclusive).

Note

Input 1 should be lesser than Input 2. Both the inputs should be positive.

Range must always be greater than zero.

If any of the condition mentioned above fails, then display ?Provide valid input?

Use a minimum of one for loop and one while loop

Sample Input 1:

2

15

Sample Output 1:

2 3 5 7 11 13

Sample Input 2:

8

5

Sample Output 2:

Provide valid input

**Source: Cognizant\Que 8\Problem Statement.txt**

Problem Statement ? Goutam and Tanul plays by telling numbers. Goutam says a number to Tanul. Tanul should first reverse the number and check if it is same as the original. If yes, Tanul should say ?Palindrome?. If not, he should say ?Not a Palindrome?. If the number is negative, print ?Invalid Input?. Help Tanul by writing a program.

Sample Input 1 :

21212

Sample Output 1 :

Palindrome

Sample Input 2 :

6186

Sample Output 2 :

Not a Palindrome

---

**Source: Cognizant\Que 9\Problem Statement.txt**

XYZ Technologies is in the process of increment the salary of the employees. This increment is done based on their salary and their performance appraisal rating.

If the appraisal rating is between 1 and 3, the increment is 10% of the salary.

If the appraisal rating is between 3.1 and 4, the increment is 25% of the salary.

If the appraisal rating is between 4.1 and 5, the increment is 30% of the salary.

Help them to do this, by writing a program that displays the incremented salary. Write a class ?IncrementCalculation.java? and write the main method in it.

Note : If either the salary is 0 or negative (or) if the appraisal rating is not in the range 1 to 5 (inclusive), then the output should be ?Invalid Input?.

Sample Input 1 :

Enter the salary  
8000

Enter the Performance appraisal rating  
3

Sample Output 1 :

8800

Sample Input 2 :

Enter the salary

7500

Enter the Performance appraisal rating

4.3

Sample Output 2 :

9750

Sample Input 3 :

Enter the salary

-5000

Enter the Performance appraisal rating

6

Sample Output 3 :

Invalid Input

---

### **Source: TATA ELXSI\Technical.txt**

Tata Elxsi technical questions for the role of Embedded Engineer and Software Engineer are given below. These questions were asked in previous Tata Elxsi technical section.

### **Tata Elxsi Technical questions - (Embedded Engineer)**

The Tata Elxsi technical questions asked for the role Embedded Engineer are given below. Students from ECE, EEE and E&I can ideally apply for this role as the Tata Elxsi technical questions will be based on their core subjects.

Check the syllabus for TATA ELXSI Embedded engineer role

1) CRO is in X-Y mode, sinusoidal inputs have 90 degrees phase difference between them. What will appear on the display?

a) circle

b) line

c) ellipse

d) triangular wave

2) Ideal Power Supply has

a) Zero internal resistance

b) High O/P resistance

c) High I/P resistance

d) Low O/P resistance

3) For the same peak value of current, which waveform will have least RMS value

a) Sine

b) Square

c) Triangular

d) Full wave rectified wave

4) A basic memory cell of dynamic RAM

a) Capacitance

b) Flip-flop

c) Transistor

d) Transistor with Capacitor

5) Ideal Power Supply has

a) Zero internal resistance

b) High O/P resistance

c) High I/P resistance

d) Low O/P resistance

6) How many comparators are required in an n-bit flash ADC?

a)  $2^{(n-1)}$

b)  $(2^n)-1$

c)  $2^n$

d) n

10) The inverter can be produced with how many NAND gates?

a) 1

b) 2

c) 3

d) 4

11) Which type of transmission line will have the maximum value of characteristic impedance ?

a) OpenWire line

b) Coaxial Cable

c) Twin lead line

d) None

12) What is an EEPROM?

a) Programmable transistor

b) Transistor

c) SRAM

d) None of these

13) The Maximum number of variables that can be represented in K-MAP?

a) 3

b) 5

c) 7

d) none of these

14) The circuits in the 8085A that provide the arithmetic and logic functions are called the:

a) CPU

b) ALU

c) I/O

d) None of these

15) Which of the following expressions is in the sum-of-products (SOP) form?

a)  $(A+B)(C+D)$

b)  $(A)B(CD)$

c)  $AB(CD)$

d)  $AB+CD$

Tata Elxsi Technical questions - (CSE and related branches)

1) Predict the output or error for the following:

```
void main()
```

```
{
```

```
int const * p=5;
```



```
printf("%d",++(*p));
```

```
}
```

A) 6

B) 5

C) Compile time error

D) None

Answer: d

Explanation:

Compiler error: Cannot modify a constant value. p is a pointer to a "constant integer". But we tried to change the value of the "constant integer".

2) Predict the output or error for the following:

```
int main()
```

```
{
```

```
char s[]="man";
```

```
int i;
```

```
for(i=0;s[ i ];i++)
```

```
printf("%c%c%c%c%c\n", s[i], (s+i), (i+s), i[s]);
```

```
return 0;
```

```
}
```

A) mmmm

aaaa

nnnn

B) mann

mann

mann

C) mmma

mma

mma

mma

D) Compile time error

Answer: a

Explanation:

$s[i]$ ,  $(i+s)$ ,  $(s+i)$ ,  $i[s]$  are all different ways of expressing the same idea. Generally,  $arrayname$  is the base address for that array. Here  $s$  is the base address.  $i$  is the index number/displacement from the base address. So, indicating it with  $*$  is same as  $s[i]$ .  $i[s]$  may be surprising. But in the case of C it is same as  $s[i]$ .

3) Predict the output or error for the following:

```
int main()
{
float me = 1.1;

double you = 1.1;

if(me==you)

printf("TATA");

else

printf("TATA Elxsi");

}
```

A) TATA

B) TATA Elxsi

C) No Output

D) Error

Answer: b

Explanation:

For floating point numbers (float, double, long double) the values cannot be predicted exactly. Depending on the number of bytes, the precision with of the value represented varies. Float takes 4 bytes and long double takes 10 bytes. So float stores 0.9 with less precision than long double.

Rule of Thumb:

Never compare or at-least be cautious when using floating point numbers with relational operators (== , >, <, <=, >=, != )  
.

4) Predict the output or error for the following:

```
int main()
{
static int var = 5;

printf("%d ",var--);

if(var)
main();

}
```

A) 5 4 3 2 1

B) 1 2 3 4 5

C) Infinite loop

D) Error

Answer: a

Explanation:

When static storage class is given, it is initialized once. The change in the value of a static variable is retained even between the function calls.

5) Predict the output or error for the following:

```
int main()
{
int c[ ]={2.8,3.4,4,6.7,5};

int j,p=c,q=c;

for(j=0;j<5;j++)

{

printf(" %d ",*c);

++q;

}

for(j=0;j<5;j++)

{

printf(" %d ",*p);

++p;

}

}
```

- A) 2 3 4 5 6 7 8 9 10
- B) 2 2 2 2 2 2 3 4 6 5
- C) Compile time error
- D) 2 3 4 6 5 2 3 4 6 5

Answer: b

Explanation:

Initially pointer c is assigned to both p and q. In the first loop, since only q is incremented and not c, the value 2 will be printed 5 times. In second loop p itself is incremented. So the values 2 3 4 6 5 will be printed.

6) Predict the output or error for the following:

```
int main()
{
extern int i;

i=20;

printf("%d", i);
}
```

- A) Error
- B) 20
- C) Garbage value
- D) 0

Answer: a

Explanation:

Linker Error: Undefined symbol '\_i'

extern storage class in the following declaration, extern int i;

specifies to the compiler that the memory for i is allocated in some other program and that address will be given to the current program at the time of linking. But linker finds that no other variable of name i is available in any other program with memory space allocated for it. Hence a linker error has occurred.

7) Predict the output or error for the following:

```
int main()
```

```
{
int i=-1, j=-1, k=0, l=2, m;

m=i++ && j++ && k++ || l++;

printf("%d %d %d %d %d", i, j, k, l, m);

}
```

- A) 0 0 1 3 1
- B) -1 -1 0 2 1
- C) -1 -1 0 2 0
- D) Error

Answer: a

Explanation:

Logical operations always give a result of 1 or 0. And also the logical AND (&&) operator has higher priority over the logical OR (||) operator. So the expression `i++ && j++ && k++` is executed first. The result of this expression is 0 (`-1 && -1 && 0 = 0`). Now the expression is `0 || 2` which evaluates to 1 (because OR operator always gives 1 except for `0 || 0` combination- for which it gives 0). So the value of `m` is 1. The values of other variables are also incremented by 1.

8) Predict the output or error for the following:

```
int main()

{

char *p;

printf("%d %d ",sizeof(*p),sizeof(p));

}
```

- A) 1 1
- B) 1 2
- C) 1 8
- D) 2 4

Answer: c

Explanation:

The size of() operator gives the number of bytes taken by its operand.

\*p refers the character stored and the size of any character is 1 byte.

Size of any pointer variable will be either 4 bytes or 8 bytes based on the compiler (32 bit or 64 bit).

9) Predict the output or error for the following:

```
int main()
{
    int i = 4;

    switch(i)
    {
        default: printf("zero");

        case 1: printf("one");

        break;

        case 2: printf("two");

        break;

        case 3: printf("three");

        break;

    }
}
```

A) zero one

B) zero

C) Compile time error

D) No output

Answer: a

Explanation:

The default case can be placed anywhere inside the loop. It is executed only when all other cases doesn't match.

10) Predict the output or error for the following:

```
int main()
{
printf("%x", -1<<4);
}
```

A)ffff

B) fff0

C) fff1

D) Error

Answer: b

Explanation :

-1 is internally represented as all 1's. When left shifted four times the least significant 4 bits are filled with 0's. The %x format specifier specifies that the integer value be printed as a hexadecimal value.

11) Predict the output or error for the following:

```
int main()
{
char string[]="Hello World";

display(string);
}
```



```
void display(char *string)

{

printf("%s", string);

}
```

A) Hello World

B) Hello

C) World

D) Error

Answer: d

Explanation:

Compiler Error: Type mismatch in re declaration of function display

In the third line, when the function display is encountered, the compiler doesn't know anything about the function display. It assumes the arguments and return types to be integers, (which is the default type). When it sees the actual function display, the arguments and type contradicts with what it has assumed previously. Hence a compile-time error occurs.

12) Predict the output or error for the following:

```
int main()

{

int c=- -2;

printf("%d", c);

}
```

A) 2

B) -2

C) 1

D) Error

Answer: a

Explanation:

Here unary minus (or negation) operator is used twice. Same maths rules applies, ie. minus \* minus = plus.

Note: However you cannot give like --2. Because -- operator can only be applied to variables as a decrement operator (eg., i--). 2 is a constant and not a variable.

13) Predict the output or error for the following:

```
#define int char

void main()

{

int i=65;

printf("%d",sizeof(i));

}
```

A) 1

B) 2

C) Error

D) Garbage Value

Answer: a

Explanation:

Since the #define replaces the string int by the macro char.

14) Predict the output or error for the following:

```
int main()
```

```
{
int i =10;

i = !i > 14;

printf ("%d", i);

}
```

- A) 1
- B) 14
- C) 0
- D) error

Answer: c

Explanation:

In the expression `!i>14` , NOT (!) operator has more precedence than > symbol. ! is a unary logical operator. `!i` (`!10`) is 0 (not of true is false). `0>14` is false (zero).

15) Predict the output or error for the following:

```
int main()

{

struct xx

{

int x=3;

char name[]="hello";

};

struct xx *s;

printf("%d", s->x);

printf("%s", s->name);
```

```
}

```

A) 3hello

B) 3h

C) Compile time error

D) Garbage value hello

Answer: c

Explanation:

You should not initialize variables in

**Source: TCS\Que 1\Problem Statement.txt**

TCS NQT Coding Question 2022 ? September Day 1 ? Slot 1

Problem Statement ? An automobile company manufactures both a two wheeler (TW) and a four wheeler (FW). A company manager wants to make the production of both types of vehicle according to the given data below:

1st data, Total number of vehicle (two-wheeler + four-wheeler)=v

2nd data, Total number of wheels = W

The task is to find how many two-wheelers as well as four-wheelers need to manufacture as per the given data.

Example :

Input :

200 -&gt; Value of V

540 -&gt; Value of W

Output :

TW =130 FW=70

Explanation:

130+70 = 200 vehicles

(70\*4)+(130\*2)= 540 wheels

Constraints :

$2 \leq W$   
 $W \% 2 = 0$   
 $V < W$

Print "INVALID INPUT" , if inputs did not meet the constraints.

The input format for testing

The candidate has to write the code to accept two positive numbers separated by a new line.

First Input line : Accept value of V.

Second Input line- Accept value for W.

The output format for testing

Written program code should generate two outputs, each separated by a single space character(see the example)  
 Additional messages in the output will result in the failure of test case

### Source: TCS\Que 10\Problem Statement.txt

#### Problem Statement

The Caesar cipher is a type of substitution cipher in which each alphabet in the plaintext or messages is shifted by a number of places down the alphabet.

For example, with a shift of 1, P would be replaced by Q, Q would become R, and so on.

To pass an encrypted message from one person to another, it is first necessary that both parties have the "Key" for the cipher, so that the sender may encrypt and the receiver may decrypt it.

Key is the number of OFFSET to shift the cipher alphabet. Key can have basic shifts from 1 to 25 positions as there are 26 total alphabets.

As we are designing custom Caesar Cipher, in addition to alphabets, we are considering numeric digits from 0 to 9. Digits can also be shifted by key places.

For Example, if a given plain text contains any digit with values 5 and key = 2, then 5 will be replaced by 7, "-"(minus sign) will remain as it is. Key value less than 0 should result into "INVALID INPUT"

Example 1:

Enter your PlainText: All the best

Enter the Key: 1

The encrypted Text is: Bmm uif Cftu

Write a function CustomCaesarCipher(int key, String message) which will accept plaintext and key as input parameters and returns its cipher text as output.

**Source: TCS\Que 11\Problem Statement.txt****Problem Statement**

We want to estimate the cost of painting a property. Interior wall painting cost is Rs.18 per sq.ft. and exterior wall painting cost is Rs.12 per sq.ft.

Take input as

1. Number of Interior walls
2. Number of Exterior walls
3. Surface Area of each Interior Wall in units of square feet
4. Surface Area of each Exterior Wall in units of square feet

If a user enters zero as the number of walls then skip Surface area values as User may don't want to paint that wall.

Calculate and display the total cost of painting the property

Example 1:

```
6
3
12.3
15.2
12.3
15.2
12.3
15.2
10.10
10.10
10.00
Total estimated Cost : 1847.4 INR
```

Note: Follow in input and output format as given in above example

**Source: TCS\Que 12\Problem Statement.txt****Problem Statement**

A City Bus is a Ring Route Bus which runs in circular fashion. That is, Bus once starts at the Source Bus Stop, halts at each Bus Stop in its Route and at the end it reaches the Source Bus Stop again.

If there are n number of Stops and if the bus starts at Bus Stop 1, then after nth Bus Stop, the next stop in the Route will be Bus Stop number 1 always.

If there are n stops, there will be n paths. One path connects two stops. Distances (in meters) for all paths in Ring Route is given in array Path[] as given below:

Path = [800, 600, 750, 900, 1400, 1200, 1100, 1500]

Fare is determined based on the distance covered from source to destination stop as Distance between Input Source and Destination Stops can be measured by looking at values in array Path[] and fare can be calculated as per following criteria:

If d = 1000 metres, then fare = 5 INR

(When calculating fare for others, the calculated fare containing any fraction value should be ceiled. For example, for

distance 900n when fare initially calculated is 4.5 which must be ceiled to 5)

Path is circular in function. Value at each index indicates distance till current stop from the previous one. And each index position can be mapped with values at same index in BusStops [] array, which is a string array holding abbreviation of names for all stops as-

?THANERAILWAYSTN? = ?TH?, ?GAONDEVI? = ?GA?, ?ICEFACTROY? = ?IC?, ?HARINIWASCIRCLE? = ?HA?,  
 ?TEENHATHNAKA? = ?TE?, ?LUISWADI? = ?LU?, ?NITINCOMPANYJUNCTION? = ?NI?, ?CADBURRYJUNCTION?  
 = ?CA?

Given, n=8, where n is number of total BusStops.

BusStops = [ ?TH?, ?GA?, ?IC?, ?HA?, ?TE?, ?LU?, ?NI?,?CA? ]

Write a code with function getFare(String Source, String Destination) which take Input as source and destination stops(in the format containing first two characters of the Name of the Bus Stop) and calculate and return travel fare.

Example 1:

Input Values

ca

Ca

Output Values

INVALID OUTPUT

Example 2:

Input Values

NI

HA

Output Values

23.0 INR

Note: Input and Output should be in format given in example.

Input should not be case sensitive and output should be in the format INR

## Source: TCS\Que 13\Problem Statement.txt

### Problem Statement

There are total n number of Monkeys sitting on the branches of a huge Tree. As travelers offer Bananas and Peanuts, the Monkeys jump down the Tree. If every Monkey can eat k Bananas and j Peanuts. If total m number of Bananas and p number of Peanuts are offered by travelers, calculate how many Monkeys remain on the Tree after some of them jumped down to eat.

At a time one Monkeys gets down and finishes eating and go to the other side of the road. The Monkey who climbed down does not climb up again after eating until the other Monkeys finish eating.

Monkey can either eat k Bananas or j Peanuts. If for last Monkey there are less than k Bananas left on the ground or less than j Peanuts left on the ground, only that Monkey can eat Bananas(<k) along with the Peanuts(<j).

Write code to take inputs as n, m, p, k, j and return the number of Monkeys left on the Tree.

Where, n= Total no of Monkeys

k= Number of eatable Bananas by Single Monkey (Monkey that jumped down last may get less than k Bananas)

j = Number of eatable Peanuts by single Monkey(Monkey that jumped down last may get less than j Peanuts)

m = Total number of Bananas

p = Total number of Peanuts

Remember that the Monkeys always eat Bananas and Peanuts, so there is no possibility of k and j having a value zero

Example 1:

Input Values

20  
2  
3  
12  
12

Output Values

Number of Monkeys left on the tree:10

Note: Kindly follow the order of inputs as n,k,j,m,p as given in the above example. And output must include the same format as in above example(Number of Monkeys left on the Tree:)

For any wrong input display INVALID INPUT

### Source: TCS\Que 14\Problem Statement.txt

#### Problem Statement

Chain Marketing Organization has has a scheme for income generation, through which its members generate income for themselves. The scheme is such that suppose A joins the scheme and makes R and V to join this scheme then A is Parent Member of R and V who are child Members. When any member joins the scheme then the parent gets total commission of 10% from each of its child members.

Child members receive commission of 5% respectively. If a Parent member does not have any member joined under him, then he gets commission of 5%.

Take name of the members joining the scheme as input.

Display how many members joined the scheme including parent member. Calculate the Total commission gained by each members in the scheme. The fixed amount for joining the scheme is Rs.5000 on which commission will be generated

SchemeAmount = 5000

Example 1: When there are more than one child members

Input : (Do not give input prompts. Accept values as follows. )

Amit //Enter parent Member as this  
Y //Enter Y if Parent member has child members otherwise enter N  
Rajesh, Virat //Enter names of child members of Amit in comma separated

Output: (Final Output must be in format given below.)

TOTAL MEMBERS:3

COMISSION DETAILS

Amit: 1000 INR

Rajesh :250 INR

Virat: 250 INR

Example 2: When there is only one child member in the hierarchy

Input :

Amit



Y

Rajesh

Output:

Total Members: 2

Comission Details

Amit: 500 INR

Rajesh: 250 INR

---

**Source: TCS\Que 15\Problem Statement.txt**

Problem Statement

FULLY AUTOMATIC VENDING MACHINE ? dispenses your cuppa on just press of button. A vending machine can serve range of products as follows:

Coffee

Espresso Coffee

Cappuccino Coffee

Latte Coffee

Tea

Plain Tea

Assam Tea

Ginger Tea

Cardamom Tea

Masala Tea

Lemon Tea

Green Tea

Organic Darjeeling Tea

Soups

Hot and Sour Soup

Veg Corn Soup

Tomato Soup

Spicy Tomato Soup

Beverages

Hot Chocolate Drink

Badam Drink

Badam-Pista Drink

Write a program to take input for main menu & sub menu and display the name of sub menu selected in the following format (enter the first letter to select main menu):

Welcome to CCD

Enjoy your

Example 1:

Input:

c

1

Output

Welcome to CCD!

Enjoy your Espresso Coffee!

Example 2:

Input

t

9

Output

INVALID OUTPUT!

### Source: TCS\Que 16\Problem Statement.txt

#### Problem Statement

A doctor has a clinic where he serves his patients. The doctor's consultation fees are different for different groups of patients depending on their age. If the patient's age is below 17, fees is 200 INR. If the patient's age is between 17 and 40, fees is 400 INR. If patient's age is above 40, fees is 300 INR. Write a code to calculate earnings in a day for which one array/List of values representing age of patients visited on that day is passed as input.

Note:

Age should not be zero or less than zero or above 120

Doctor consults a maximum of 20 patients a day

Enter age value (press Enter without a value to stop):

Example 1:

Input

20

30

40

50

2

3

14

Output

Total Income 2000 INR

Note: Input and Output Format should be same as given in the above example.

For any wrong input display INVALID INPUT

Output Format

Total Income 2100 INR

**Source: TCS\Que 17\Problem Statement.txt**

Explanation:

To check whether a year is leap or not

Step 1:

We first divide the year by 4.

If it is not divisible by 4 then it is not a leap year.

If it is divisible by 4 leaving remainder 0

Step 2:

We divide the year by 100

If it is not divisible by 100 then it is a leap year.

If it is divisible by 100 leaving remainder 0

Step 3:

We divide the year by 400

If it is not divisible by 400 then it is a leap year.

If it is divisible by 400 leaving remainder 0

Then it is a leap year

**Source: TCS\Que 18\Problem Statement.txt**

Ques. Write a code to check whether no is prime or not. Condition use function check() to find whether entered no is positive or negative ,if negative then enter the no, And if yes pas no as a parameter to prime() and check whether no is prime or not?

Whether the number is positive or not, if it is negative then print the message ?please enter the positive number?

It is positive then call the function prime and check whether the take positive number is prime or not.

**Source: TCS\Que 19\Problem Statement.txt**

Find the 15th term of the series?

0,0,7,6,14,12,21,18, 28

Explanation : In this series the odd term is increment of 7 {0, 7, 14, 21, 28, 35 ? ? ? ? ? }

And even term is a increment of 6 {0, 6, 12, 18, 24, 30 ? ? ? ? ? }

**Source: TCS\Que 2\Problem Statement.txt**

TCS NQT Coding Question 2022 ? September Day 1 ? Slot 1

Problem Statement ? Given a string S(input consisting) of ?? and ?#?. The length of the string is variable. The task is to find the minimum number of ?? or ?#? to make it a valid string. The string is considered valid if the number of ?? and

?? are equal. The ?? and ?#? can be at any position in the string.

Note : The output will be a positive or negative integer based on number of ?? and ?#? in the input string.

(\*>#): positive integer

(>#): negative integer

(>#): 0

Example 1:

Input 1:

###\* -> Value of S

Output :

0 ? number of \* and are equal

#### Source: TCS\Que 20\Problem Statement.txt

Consider the following series: 1, 1, 2, 3, 4, 9, 8, 27, 16, 81, 32, 243, 64, 729, 128, 2187 ?

This series is a mixture of 2 series ? all the odd terms in this series form a geometric series and all the even terms form yet another geometric series. Write a program to find the Nth term in the series.

The value N in a positive integer that should be read from STDIN. The Nth term that is calculated by the program should be written to STDOUT. Other than value of n th term, no other character / string or message should be written to STDOUT. For example , if N=16, the 16th term in the series is 2187, so only value 2187 should be printed to STDOUT.

You can assume that N will not exceed 30.

#### Source: TCS\Que 21\Problem Statement.txt

Consider the below series :

0, 0, 2, 1, 4, 2, 6, 3, 8, 4, 10, 5, 12, 6, 14, 7, 16, 8

This series is a mixture of 2 series all the odd terms in this series form even numbers in ascending order and every even terms is derived from the previous term using the formula  $(x/2)$

Write a program to find the nth term in this series.

The value n in a positive integer that should be read from STDIN the nth term that is calculated by the program should be written to STDOUT. Other than the value of the nth term no other characters /strings or message should be written to STDOUT.

For example if  $n=10$ , the 10th term in the series is to be derived from the 9th term in the series. The 9th term is 8 so the 10th term is  $(8/2)=4$ . Only the value 4 should be printed to STDOUT.

You can assume that the  $n$  will not exceed 20,000.

**Source: TCS\Que 22\Problem Statement.txt**

1. The program will receive 3 English words inputs from STDIN

These three words will be read one at a time, in three separate line

The first word should be changed like all vowels should be replaced by %

The second word should be changed like all consonants should be replaced by The third word should be changed like all char should be converted to upper case

Then concatenate the three words and print them

Other than these concatenated word, no other characters/string should or message should be written to STDOUT

For example if you print how are you then output should be h%wa#eYOU.

You can assume that input of each word will not exceed more than 5 chars

**Source: TCS\Que 23\Problem Statement.txt**

1. Using a method, pass two variables and find the sum of two numbers.

Test case:

Number 1 ? 20

Number 2 ? 20.38

Sum = 40.38

There were a total of 4 test cases. Once you compile 3 of them will be shown to you and 1 will be a hidden one. You have to display error message if numbers are not numeric.

**Source: TCS\Que 24\Problem Statement.txt**

Consider the below series :

0, 0, 2, 1, 4, 2, 6, 3, 8, 4, 10, 5, 12, 6, 14, 7, 16, 8

This series is a mixture of 2 series all the odd terms in this series form even numbers in ascending order and every even terms is derived from the previous term using the formula  $(x/2)$

Write a program to find the  $n$ th term in this series.

The value  $n$  is a positive integer that should be read from STDIN the  $n$ th term that is calculated by the program should

be written to STDOUT. Other than the value of the nth term no other characters /strings or message should be written to STDOUT.

For example if  $n=10$ , the 10th term in the series is to be derived from the 9th term in the series. The 9th term is 8 so the 10th term is  $(8/2)=4$ . Only the value 4 should be printed to STDOUT.

You can assume that the  $n$  will not exceed 20,000.

### Source: TCS\Que 3\Problem Statement.txt

TCS Coding Question Day 1 Slot 2 ? Question 1

Given an integer array Arr of size N the task is to find the count of elements whose value is greater than all of its prior elements.

Note : 1st element of the array should be considered in the count of the result.

For example,

$Arr[] = \{7, 4, 8, 2, 9\}$

As 7 is the first element, it will consider in the result.

8 and 9 are also the elements that are greater than all of its previous elements.

Since total of 3 elements is present in the array that meets the condition.

Hence the output = 3.

Example 1:

Input

5 -> Value of N, represents size of Arr

7-> Value of Arr[0]

4 -> Value of Arr[1]

8-> Value of Arr[2]

2-> Value of Arr[3]

9-> Value of Arr[4]

Output :

3

Example 2:

5 -> Value of N, represents size of Arr

3 -> Value of Arr[0]

4 -> Value of Arr[1]

5 -> Value of Arr[2]

8 -> Value of Arr[3]

9 -> Value of Arr[4]

Output :

5

Constraints

$1 \leq N \leq 20$

$1 \leq \text{Arr}[i] \leq 10000$

### Source: TCS\Que 4\Problem Statement.txt

TCS Coding Question Day 1 Slot 2 ? Question 2

A parking lot in a mall has  $R \times C$  number of parking spaces. Each parking space will either be empty(0) or full(1). The status (0/1) of a parking space is represented as the element of the matrix. The task is to find index of the prpeinzta row(R) in the parking lot that has the most of the parking spaces full(1).

Note :

$R \times C$ - Size of the matrix

Elements of the matrix M should be only 0 or 1.

Example 1:

Input :

3 -> Value of R(row)

3 -> value of C(column)

[0 1 0 1 1 0 1 1 1] -> Elements of the array  $M[R][C]$  where each element is separated by new line.

Output :

3 -> Row 3 has maximum number of 1?s

Example 2:

input :

4 -> Value of R(row)

3 -> Value of C(column)

[0 1 0 1 1 0 1 0 1 1 1 1] -> Elements of the array M[R][C]

Output :

4 -> Row 4 has maximum number of 1's

### Source: TCS\Que 5\Problem Statement.txt

TCS Coding Question Day 2 Slot 1 ? Question 1

A party has been organised on cruise. The party is organised for a limited time(T). The number of guests entering (E[i]) and leaving (L[i]) the party at every hour is represented as elements of the array. The task is to find the maximum number of guests present on the cruise at any given instance within T hours.

Example 1:

Input :

5 -> Value of T

[7,0,5,1,3] -> E[], Element of E[0] to E[N-1], where input each element is separated by new line

[1,2,1,3,4] -> L[], Element of L[0] to L[N-1], while input each element is separate by new line.

Output :

8 -> Maximum number of guests on cruise at an instance.

Explanation:

1st hour:

Entry : 7 Exit: 1

No. of guests on ship : 6

2nd hour :

Entry : 0 Exit : 2

No. of guests on ship : 6-2=4

Hour 3:

Entry: 5 Exit: 1



No. of guests on ship :  $4+5-1=8$

Hour 4:

Entry : 1 Exit : 3

No. of guests on ship :  $8+1-3=6$

Hour 5:

Entry : 3 Exit: 4

No. of guests on ship:  $6+3-4=5$

Hence, the maximum number of guests within 5 hours is 8.

Example 2:

Input:

4 -> Value of T

[3,5,2,0] -> E[], Element of E[0] to E[N-1], where input each element is separated by new line.

[0,2,4,4] -> L[], Element of L[0] to L[N-1], while input each element in separated by new line

Output:

6

Cruise at an instance

Explanation:

Hour 1:

Entry: 3 Exit: 0

No. of guests on ship: 3

Hour 2:

Entry : 5 Exit : 2

No. of guest on ship:  $3+5-2=6$

Hour 3:

Entry : 2 Exit: 4

No. of guests on ship:  $6+2-4= 4$

Hour 4:

Entry: 0 Exit : 4

No. of guests on ship :  $4+0-4=0$

Hence, the maximum number of guests within 5 hours is 6.

The input format for testing

The candidate has to write the code to accept 3 input.

First input- Accept value for number of T(Positive integer number)

Second input- Accept T number of values, where each value is separated by a new line.

Third input- Accept T number of values, where each value is separated by a new line.

The output format for testing

The output should be a positive integer number or a message as given in the problem statement(Check the output in Example 1 and Example 2)

Constraints:

$1 \leq T \leq 25$

$0 \leq E[i] \leq 500$

$0 \leq L[i] \leq 500$

### Source: TCS\Que 6\Problem Statement.txt

TCS Coding Question Day 2 Slot 1 ? Question 2

At a fun fair, a street vendor is selling different colours of balloons. He sells N number of different colours of balloons (B[]). The task is to find the colour (odd) of the balloon which is present odd number of times in the bunch of balloons.

Note: If there is more than one colour which is odd in number, then the first colour in the array which is present odd number of times is displayed. The colours of the balloons can all be either upper case or lower case in the array. If all the inputs are even in number, display the message ?All are even?.

Example 1:

7 -> Value of N

[r,g,b,b,g,y,y] -> B[] Elements B[0] to B[N-1], where each input element is separated by a new line.

Output :

r -> [r,g,b,b,g,y,y] -> r colour balloon is present odd number of times in the bunch.

Explanation:

From the input array above:

r: 1 balloon

g: 2 balloons

b: 2 balloons

y : 2 balloons

Hence , r is only the balloon which is odd in number.

Example 2:

Input:

10 -> Value of N

[a,b,b,b,c,c,c,a,f,c] -> B[], elements B[0] to B[N-1] where input each element is separated by new line.

Output :

b-> ?b? colour balloon is present odd number of times in the bunch.

Explanation:

From the input array above:

a: 2 balloons

b: 3 balloons

c: 4 balloons

f: 1 balloons

Here, both ?b? and ?f? have odd number of balloons. But ?b? colour balloon occurs first.

Hence , b is the output.

Input Format for testing

The candidate has to write the code to accept: 2 input

First input: Accept value for number of N(Positive integer number).

Second Input : Accept N number of character values (B[]), where each value is separated by a new line.

Output format for testing

The output should be a single literal (Check the output in example 1 and example 2)

Constraints:

$3 \leq N \leq 50$

$B[i] = \{a-z\} \text{ or } \{A-Z\}$

### Source: TCS\Que 7\Problem Statement.txt

There is a JAR full of candies for sale at a mall counter. JAR has the capacity N, that is JAR can contain maximum N candies when JAR is full. At any point of time. JAR can have M number of Candies where  $M \leq N$ . Candies are served to

the customers. JAR is never remain empty as when last  $k$  candies are left. JAR is refilled with new candies in such a way that JAR get full.

Write a code to implement above scenario. Display JAR at counter with available number of candies. Input should be the number of candies one customer can order at point of time. Update the JAR after each purchase and display JAR at Counter.

Output should give number of Candies sold and updated number of Candies in JAR.

If Input is more than candies in JAR, return: ?INVALID INPUT?

Given,

$N=10$ , where  $N$  is NUMBER OF CANDIES AVAILABLE

$K \leq 5$ , where  $k$  is number of minimum candies that must be inside JAR ever.

Example 1: ( $N = 10$ ,  $k \leq 5$ )

Input Value

3

Output Value

NUMBER OF CANDIES SOLD : 3

NUMBER OF CANDIES AVAILABLE : 7

Example : ( $N=10$ ,  $k \leq 5$ )

Input Value

0

Output Value

INVALID INPUT

NUMBER OF CANDIES LEFT : 10

### Source: TCS\Que 8\Problem Statement.txt

Selection of MPSC exams include a fitness test which is conducted on ground. There will be a batch of 3 trainees, appearing for running test in track for 3 rounds. You need to record their oxygen level after every round. After trainee are finished with all rounds, calculate for each trainee his average oxygen level over the 3 rounds and select one with highest oxygen level as the most fit trainee. If more than one trainee attains the same highest average level, they all need to be selected.

Display the most fit trainee (or trainees) and the highest average oxygen level.

Note:

The oxygen value entered should not be accepted if it is not in the range between 1 and 100.

If the calculated maximum average oxygen value of trainees is below 70 then declare the trainees as unfit with meaningful message as ?All trainees are unfit.

Average Oxygen Values should be rounded.

Example 1:

## INPUT VALUES

95

92

95

92

90

92

90

92

90

## OUTPUT VALUES

Trainee Number : 1

Trainee Number : 3

Note:

Input should be 9 integer values representing oxygen levels entered in order as

Round 1

Oxygen value of trainee 1

Oxygen value of trainee 2

Oxygen value of trainee 3

Round 2

Oxygen value of trainee 1

Oxygen value of trainee 2

Oxygen value of trainee 3

Round 3

Oxygen value of trainee 1

Oxygen value of trainee 2

Oxygen value of trainee 3

Output must be in given format as in above example. For any wrong input final output should display ?INVALID INPUT?

---

**Source: TCS\Que 9\Problem Statement.txt**

A washing machine works on the principle of Fuzzy System, the weight of clothes put inside it for washing is uncertain But based on weight measured by sensors, it decides time and water level which can be changed by menus given on the machine control area.

For low level water, the time estimate is 25 minutes, where approximately weight is between 2000 grams or any nonzero positive number below that.

For medium level water, the time estimate is 35 minutes, where approximately weight is between 2001 grams and 4000 grams.

For high level water, the time estimate is 45 minutes, where approximately weight is above 4000 grams.

Assume the capacity of machine is maximum 7000 grams

Where approximately weight is zero, time estimate is 0 minutes.

Write a function which takes a numeric weight in the range [0,7000] as input and produces estimated time as output is: ?OVERLOADED?, and for all other inputs, the output statement is

?INVALID INPUT?.

Input should be in the form of integer value ?

Output must have the following format ?

Time Estimated: Minutes

Example:

Input value  
2000

Output value  
Time Estimated: 25 minutes

Solution

### Source: TCS - NINJA\Que 1\Problem Statement.txt

One programming language has the following keywords that cannot be used as identifiers:

break, case, continue, default, defer, else, for, func, goto, if, map, range, return, struct, type, var

Write a program to find if the given word is a keyword or not

Test cases

Case 1

Input ? defer

Expected Output ? defer is a keyword

Case 2

Input ? While

Expected Output ? while is not a keyword

**Source: TCS - NINJA\Que 2\Problem Statement.txt**

Given a maximum of four digit to the base 17 (10 ? A, 11 ? B, 12 ? C, 13 ? D ? 16 ? G} as input, output its decimal value.

Test Cases

Case 1

Input ? 1A

Expected Output ? 27

Case 2

Input ? 23GF

Expected Output ? 10980

---

**Source: TCS - NINJA\Que 3\Problem Statement.txt**

Given a maximum of 100 digit numbers as input, find the difference between the sum of odd and even position digits

Test Cases

Case 1

Input: 4567

Expected Output: 2

Explanation : Odd positions are 4 and 6 as they are pos: 1 and pos: 3, both have sum 10. Similarly, 5 and 7 are at even positions pos: 2 and pos: 4 with sum 12. Thus, difference is  $12 - 10 = 2$

Case 2

Input: 5476

Expected Output: 2

Case 3

Input: 9834698765123

Expected Output: 1

---

**Source: TCS - NINJA\Que 4\Problem Statement.txt**

Ques. Write a code to check whether no is prime or not. Condition use function check() to find whether entered no is positive or negative ,if negative then enter the no, And if yes pas no as a parameter to prime() and check whether no is prime or not?

---

**Source: TCS - NINJA\Que 5\Problem Statement.txt**

Program to check if a year is Leap Year or not ?

---

**Source: Wipro Elite NLTH\Que 1\Problem Statement.txt**

Prepsters Wipro NLTH Coding Question -1 :

NOTE:- Please comment down the code in other languages as well below ?

Ques: C Program to check if two given matrices are identical

### Source: Wipro Elite NLTH\Que 10 - Matrix Multiply\Problem Statement.txt

Matrix Multiplication in Java

We can multiply two matrices in java using binary \* operator. In case of matrix multiplication, one row element of first matrix is multiplied by all columns of second matrix. In this article, we will discuss about the multiplication of matrix in java

Matrix Multiplication in java

Matrix Multiplication in Java

Obtaining a single matrix from the entries of two matrices by using a binary operation is known as Matrix multiplication. In simpler terms, if two matrices matrix1 and matrix2 of order ab and bc are multiplied, the matrix obtained is of the order a\*c.

First Matrix :

```
1 1 1
2 2 2
3 3 3
4 4 4
```

Second Matrix :

```
1 1 1 1
2 2 2 2
3 3 3 3
```

Resultant Matrix:

```
6 6 6 6
12 12 12 12
18 18 18 18
24 24 24 24
```

Approach

Take the two matrices as input to be multiplies.  
Create the new matrix mat3 to store the product of two matrices.  
Traverse each element of the two metrices and multiply them.  
Store this product in the new matrix at the corresponding index.  
Print the Resultant matrix.

### Source: Wipro Elite NLTH\Que 11 - Remove Vowels\Problem Statement.txt

Remove Vowels from a String

In this article we will learn how to code a Java program to remove vowels from a string. ?A?, ?E?, ?I?, ?O?, ?U? are five vowels out of 26 characters in English alphabet letters. Java programming is case sensitive, and hence lowercase and uppercase characters are considered differently, so we will have to check for both the cases and then we can remove



the vowels from a string

Remove the vowels from a String in java

Approach

Step 1: Take input str as a String.

Step 2: Convert the string into character array.

Step 3: Traverse the complete array and check if the character present at particular index is vowel or not.

Step 4: If the character is vowel then continue the loop else concatenate the character into resultant string.

Step 5: Print the resultant string.

### Source: Wipro Elite NLTH\Que 12 - Odd digit sum\Problem Statement.txt

Sum of all odd digits of a Number

Java program to print sum of all odd digits of a larger number is being explained in this article. Sum of all odd digits can be found by separating each digit from the number and checking whether the digit is odd or not, if odd then add that digit.

Sample Input : 1234567

Sample Output : 16

Sum of odd digits of a number in java

Approach

Step 1: Take a long number as input from the user.

Step 2: Extract a single digit from a number.

Step 3: Check if digit is odd or not.

Step 4: If digit is odd, then add the digit to variable sum.

Step 5: Print the sum.

### Source: Wipro Elite NLTH\Que 13 - Transpose\Problem Statement.txt

Transpose of a Matrix

In this article, we will discuss the java program to print the transpose of a matrix. Converting rows of matrix into columns and columns of a matrix into rows is called transpose of a matrix. In other words, transpose of matrix $[[i][j]]$  is obtained by changing matrix $[[i][j]]$  to matrix $[[j][i]]$ .

Transpose of matrix in java

Transpose of a Matrix in Java

A transpose of a matrix is the matrix flipped over its diagonal i.e. the row and column indices of the matrix are switched. An example of this is given as follows ?

Matrix =

1 2 3

4 5 6

7 8 9

Transpose of Matrix =

1 4 7

2 5 8

3 6 9

**Source: Wipro Elite NLTH\Que 14 - Largest element 2D Array\Problem Statement.txt**

Display Largest Element in 2-D Array

In this article, we will discuss about largest element in each row in 2-D Array. Array has n-rows and m-columns , here our task is to write a program to Display the largest element in each row in a 2-D Array in Java.

Sample input: a=[[1,2,3],[6,5,4],[8,9,7]]

Sample output: 3 6 9

Largest element in 2-D Array in java

Approach

Take the row and column of a matrix as input.

Take row\*column space separated elements as input.

Declare max variable and initialize it with the Minimum value of Integer.

Traverse each element of a row and print it.

At the end all the max element of each row get printed.

**Source: Wipro Elite NLTH\Que 15 - Balanced Parenthesis\Problem Statement.txt**

Balanced Parenthesis

To check balanced parenthesis is a basic interview question where we are asked to find whether the given string(of brackets) is balanced or not. To do this , the traditional way of doing is using stacks but we can also find by using normal programming techniques. Different brackets are ( ) , [ ] , { }. Question can be asked on any type of bracket or of all types of brackets.

Sample input: ()()

Sample output: True

Balanced Parenthesis in Java

Algorithm

Declare a character stack.

Now traverse the expression string exp.

If the current character is a starting bracket ( ( or { or [ ) then push it to stack.

If the current character is a closing bracket ( ) or } or ] ) then pop from stack and if the popped character is the matching starting bracket then fine else brackets are not balanced.

After complete traversal, if there is some starting bracket left in stack then ?not balanced?

**Source: Wipro Elite NLTH\Que 16\Problem Statement.txt**

Wipro?s client eBay wants to run a campaign on their website, which will have the following parameters, eBay wants that on certain x products, they want to calculate the final price, for each product on eBay there will be a stock unit parameter, this parameter will denote, how many items are their in their fulfillment center

Now, while these numbers if are positive means product x is available in the fulfillment center and if not than the product is not available and cannot be shipped to the customer

Now the price on for each product varies based on the distance of the customer from the fulfillment center. Now, each product is in different fulfillment zone. Now, these values are 00?s kms for each centurion km. The price available would further increase by factor distance.

You?ve to find the maximum discount price for each product if the product can be shipped.

Following are the input/output parameters :

Input

The first line of the input will contain number of products.  
 The second line will contain price for each of these products.  
 The third line contains shipping distance in 00?s kms  
 The fourth line contains SKU?s  
 Output

It will contain the final price for each deliverable item in SKU?s  
 Example :

Input:

6  
 87 103 229 41 8 86  
 3 1 9 2 1 2  
 7 -21 30 0 -4 -3  
 Output

261 2061

### Source: Wipro Elite NLTH\Que 17\Problem Statement.txt

Problem: in this first line you are required to take the value of m and n as the number of rows and columns of matrix, then you are required to take the input elements of array.

As an output you are required to print the sum of each row then the row having the maximum sum.

Test Case :

Input : 3 3

1 2 3

4 5 6

7 8 9

Output :

Row 1 : 6

Row 2 : 15

Row 3 : 24

Row 3 is having the maximum sum : 24

### Source: Wipro Elite NLTH\Que 18\Problem Statement.txt

Problem: You are required to count the number of words in a sentence.

You are required to pass all the test cases

Test Cases :

Test Case : 1

Input : welcome to the world  
Output : 4

Test Case : 2  
Input : [space] say hello  
Output : 2

Test Case : 3  
Input : To get pass you need to study hard [space] [space]  
Output : 8

### Source: Wipro Elite NLTH\Que 19\Problem Statement.txt

Problem: First of all you need to input a whole sentence in the string, then you need to enter the target word.

As an output you need to find the number of times that particular target word is repeated in the sentence.

Test Cases :

Test Case : 1  
Input : welcome world to the new world  
world  
Output : 2

Test Case : 2  
Input : working hard and working smart both are different ways of working  
working  
Output : 3

### Source: Wipro Elite NLTH\Que 2\Problem Statement.txt

PrepSter Wipro NLTH Coding Question ? 2  
Print a given matrix in spiral form

Ques: Given a 2D array, print it in spiral form. See the following examples.

NOTE:- Please comment down the code in other languages as well below .

Input:

```
1  2  3  4
5  6  7  8
9 10 11 12
13 14 15 16
```

Output:

```
1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10
```

Input:

```
1  2  3  4  5  6
```

```

7 8 9 10 11 12
13 14 15 16 17 18

```

Output:

```
1 2 3 4 5 6 12 18 17 16 15 14 13 7 8 9 10 11
```

### Source: Wipro Elite NLTH\Que 20\Problem Statement.txt

#### Problem Statement

You are required to implement the following function:

```
Int SumNumberDivisible(int m, int n);
```

The function accepts 2 positive integer  $m$  and  $n$  as its arguments.

You are required to calculate the sum of numbers divisible both by 3 and 5, between  $m$  and  $n$  both inclusive and return the same.

Note

$0 < m \leq n$

Example

Input:

$m : 12$

$n : 50$

Output 90

Explanation:

The numbers divisible by both 3 and 5, between 12 and 50 both inclusive are {15, 30, 45} and their sum is 90.

Sample Input

$m : 100$

$n : 160$

Sample Output

510

### Source: Wipro Elite NLTH\Que 21\Problem Statement.txt

Input

The first line of input consists of three space separated integers. Num, start and end representing the size of list [N], the starting value of the range and the ending value of the range respectively.

The second line of input consists N space separated integers representing the distances of the employee from the company.

Output

Print space separated integers representing the ID's of the employee whose distance lie within the given range else

return -1.

Example

Input

6 30 50

29 38 12 48 39 55

Output

1 3 4

Explanation :

There are 3 employees with id 1, 3, 4 whose distance from the office lies within the given range.

### Source: Wipro Elite NLTH\Que 22\Problem Statement.txt

Write a code for the following functions

Function 1 : sortArray

It's function is to sort the passed array with a particular length, there are 2 parameters passed into it first is the length of array and next is the elements of list.

Function 2 : findMaxElement

It's function is to get the 2 array with their own particular elements and length, and find the maximum element from both the array and print it.

Input

12

2 5 1 3 9 8 4 6 5 2 3 11

11

11 13 2 4 15 17 67 44 2 100 23

Output

100

### Source: Wipro Elite NLTH\Que 23\Problem Statement.txt

Problem: you are given a number, and you have to extract the key by finding the difference between the sum of the even and odd numbers of the input.

Test Case :

Input : 24319587

Output : 11

Explanation : odd terms :  $3 + 1 + 9 + 5 + 7 = 25$

even terms :  $2 + 4 + 8 = 14$

output : 11 (25-14)

**Source: Wipro Elite NLTH\Que 24\Problem Statement.txt**

Problem: in this first line you are required to take the value of m and n as the number of rows and columns of matrix, then you are required to take the input elements of array.

As an output you are required to print the sum of each row then the row having the maximum sum.

Test Case :

Input : 3 3

1 2 3

4 5 6

7 8 9

Output :

Row 1 : 6

Row 2 : 15

Row 3 : 24

Row 3 is having the maximum sum : 24

---

**Source: Wipro Elite NLTH\Que 25\Problem Statement.txt**

Problem: in this first line you are required to take the input that set the number of elements to be inserted, the next line takes the elements as input.

You are required to print the sum of maximum and minimum element of the list

Test Case :

Input : 6

55 87 46 21 34 79

Output :

108

Explanation :

$21 + 87 = 108$

---

**Source: Wipro Elite NLTH\Que 26\Problem Statement.txt**

Problem: in this first line you are required to take the input of an integer number, and in the second line you are required to input the target element

You are required to print the number of time target element occurred in the integer

Test Case :

Input : 734139

3

Output :

2

Explanation :

3 occurred 2 times in the integer.

### Source: Wipro Elite NLTH\Que 3\Problem Statement.txt

PrepSter Wipro NLTH Coding Question ? 3

Ques: Given an n-by-n matrix of 0's and 1's where all 1's in each row come before all 0's, find the most efficient way to return the row with the maximum number of 0's.

{1,1,1,1},

{1,1,0,0},

{1,0,0,0},

{1,1,0,0},

NOTE:- Please comment down the code in other languages as well below ?

### Source: Wipro Elite NLTH\Que 4\Problem Statement.txt

PrepSter Wipro NLTH Coding Question ? 4

Ques: A Pythagorean triplet is a set of three integers a, b and c such that  $a^2 + b^2 = c^2$ . Given a limit, generate all Pythagorean Triples with values smaller than given limit.

Input : limit = 20

Output : 3 4 5

8 6 10

5 12 13

15 8 17

12 16 20

A Simple Solution is to generate these triplets smaller than given limit using three nested loop. For every triplet, check if Pythagorean condition is true, if true, then print the triplet. Time complexity of this solution is  $O(\text{limit}^3)$  where ?limit? is given limit.



An Efficient Solution can print all triplets in  $O(k)$  time where  $k$  is number of triplets printed. The idea is to use square sum relation of Pythagorean triplet, i.e., addition of squares of  $a$  and  $b$  is equal to square of  $c$ , we can write these number in terms of  $m$  and  $n$  such that,

$$a = m^2 - n^2$$

$$b = 2 \cdot m \cdot n$$

$$c = m^2 + n^2$$

because,

$$a^2 = m^4 + n^4 - 2 \cdot m^2 \cdot n^2$$

$$b^2 = 4 \cdot m^2 \cdot n^2$$

$$c^2 = m^4 + n^4 + 2 \cdot m^2 \cdot n^2$$

We can see that  $a^2 + b^2 = c^2$ , so instead of iterating for  $a$ ,  $b$  and  $c$  we can iterate for  $m$  and  $n$  and can generate these triplets

### Source: Wipro Elite NLTH\Que 5 - GCD\Problem Statement.txt

Program to calculate GCD of Two Numbers

In this article, we will discuss about GCD of Two Numbers in Java. GCD of Two Numbers or integers (where integers not equal to zero) is largest positive integer that divides each of the integers. For example, GCD of 12 and 16 is 4, where factors of 12==>1,2,4,6,12

factors of 16==>1,2,4,8,16

common factors are 1,2,4

greatest common factor is 4.

Program to check a number is prime or not in C++

Algorithm for GCD of Two Numbers:

Take Two Numbers num1 and num2 as input.

Initialize a variable  $i$  to minimum of num1 and num2 and loop until  $i$  is greater than or equal to 1.

Check if  $i$  divides num1 and num2 completely or not. If divides completely then break the loop.

Now, print the value of  $i$ .

### Source: Wipro Elite NLTH\Que 6 - GCD of 3 no\Problem Statement.txt

Program to calculate GCD of three numbers

In this article we will discuss about GCD of three numbers. GCD of three integers (where integers not equal to zero) is largest positive integer that divides each of the three integers. For example GCD of 12, 16, 22 is 2 where factors of 12==>1,2,3,4,6,12

factors of 16==>1,2,4,8,16

factors of 22==>1,2,11,22

common factors==>1,2

greatest common factor==>2

GCD of three numbers in java

Algorithm for GCD of three numbers:

Take Three Numbers num1, num2 and num3 as input.

Initialize a variable  $i$  to minimum of num1, num2, num3 and loop until  $i$  is greater than or equal to 1.

Check if  $i$  divides num1, num2, num3 completely or not. If divides completely then break the loop.

Now, print the value of  $i$ .

**Source: Wipro Elite NLTH\Que 7 - Prime\Problem Statement.txt**

Check if Number is Prime or Not

A number is said to be Prime Number if and only if it is divisible by one and itself. In this article, we will discuss about the program to check if the given number is prime or not.

Sample Input : 17

Sample Output : 17 is a Prime Number

Prime Number in java

Steps to Check if the number is Prime or not

Step 1: Take input as a function parameter.

Step 2: Check if the number is less than to 1 or not. If it is less than 1, return false.

Step 3: Check if no is less than or equal to 3 or not. If it is less than or equal to 3, return true.

Step 4: Now, check if no is completely divisible by 2 or 3 or not. If it is complete divisible by 2 or 3 then return false.

Step 5: Now, initialize variable i to 5 and loop until i\*i is less than or equal to or not and every time increment i with 6.

Step 6: Check if the number if completely divisible by i or i+2 or not. If it is completely divisible then return false.

Step 7: If the control reach to end then return true.

**Source: Wipro Elite NLTH\Que 8 - Distinct elements of Array\Problem Statement.txt**

Distinct Elements of an Array in Java

Distinct elements of are nothing but a unique (non-duplicate) elements present in the given array. In this article, we will discuss how to find the distinct elements of an array in java.

Sample Input : 1, 3, 7 11, 3, 8, 11

Sample Output : 1, 3, 7, 11, 8

Distinct elements of an array in java

Steps to find distinct elements of an array

Take the size of array as input n.

Take n space separated elements of an array as input.

Traverse the array from the beginning.

Check if the current element is found in the array again.

If it is found, then do not print that element.

Else, print that element and continue.

**Source: Wipro Elite NLTH\Que 9 - Sort Half\Problem Statement.txt**

Sort array ascending and descending order in java

In this article we will learn how to write a java program for sorting first half in ascending order and second half in descending order. Although it sounds a bit difficult but we'll try to make it as easy as possible. Let's take an example for understanding what we have to perform

Input array : [3, 5, 2, 7, 9, 1]

Output array : [1, 2, 3, 9, 7, 5]

Algorithm for Sorting first half in ascending order and second half in descending order (Sort array ascending and descending order)

Step 1 ? Initialize a variable len, for accepting the length of the array.

Step 2 ? Take the input from the user using a for loop.

Step 3 ? Sort the inputted array in the ascending order.

Step 4 ? Use two for loops for printing the array

Use the first loop from 0 to len/2 for printing the first ascending part of the array.

Use the second for loop from len-1 to len/2 for printing the second descending part of the array.

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